

Local Search with Correlated Randomness

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Abstract

How much does an algorithm’s running-time distribution under independent randomness reveal about its behavior when independence is no longer guaranteed? We study sources satisfying $\nu[w] \leq DP[w]^s$ for every finite prefix w , where P is an independent reference law, $0 < s \leq 1$, and $D \geq 1$. The constraint controls complete-prefix probabilities while allowing individual choices to be predictable, even fully determined by the past.

For retry tasks, all deterministic history-dependent selectors have the same independent-source running-time law. Yet two orders have worst-case failure probabilities 1 and $\exp[-\Theta(n)]$ at the same linear deadline under the same source constraint. We identify a static priority rule that is optimal at every deadline and every D .

For the standard local walk on a k -CNF with at least r true literals per clause under some assignment, $k/2 < r < k$, we determine the sharp source threshold s_* . At and above it, the expected flip count is $O_{k,r}(\min\{L^3, L/(s - s_*)\})$, where $L = h + \log D + 1$, h is the initial Hamming distance to that assignment, and $L/0 = \infty$. The bound allows arbitrary clause overlap and history-dependent clause selection. Matching instances admit one source forcing this delay with probability one for every selector. At criticality and fixed D , the delay is cubic despite a linear independent-source expectation.

Variable-depth prefix covers, together with classical tree max-flow/min-cut, yield an exact criterion for restoring exponential tails by restarting on the same tape. We synthesize updates and restarts for explicit finite-state processes. Under a sufficient prefix guarantee, we also obtain noisy predecessor search with error at most η and expected query count polynomial in the correct leaf’s depth and $\log(D/\eta)$, without knowing the depth or tree height.

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1 Introduction

How much does an algorithm’s running-time distribution under independent randomness reveal about its behavior when independence is no longer guaranteed? The distribution records the probability of surviving to each time, but not which early choices the surviving runs share. When later choices can depend on earlier ones, that missing structure matters. We give retry tasks whose orders have identical independent-source running-time laws, yet under the same bound on prefix probabilities, two orders miss a linear deadline with worst-case probabilities 1 and $\exp[-\Theta(n)]$. We then identify an order that is optimal at every deadline, even against history-dependent selection of tasks.

What randomness does local search need? Tompkins and Hoos [TH06] examined how randomness affects SAT local search. They observed robustness to derandomization, but also failures caused by cyclic streams. We study this issue under a quantitative guarantee on the randomness consumed by the algorithm. Consider a 3-CNF with an assignment making at least two literals true in every clause. Select a violated clause and use an index from $\{1, 2, 3\}$ to choose a variable to flip. At least two choices move toward the promised assignment, although the walk does not know it. Initialization and clause selection can be deterministic.

Repeating one index for m steps. For a positive integer m , generate $18m$ independent uniform indices in $\{1, 2, 3\}$, repeat the last index m times, then generate independent uniform indices forever. At every fixed position, each index has probability $1/3$. Once the first $18m$ indices have been observed, however, the next m indices are known. Every possible prefix of length t fixes at least $18t/19$ independent draws, so its probability is at most $3^{-18t/19}$. Our SAT theorem guarantees termination for this source because $18/19 > \log_3(2\sqrt{2}) \approx 0.94639$, the threshold for the walk above. The walk uses each index as it arrives. A restart during the repeated block changes the assignment, but the next index remains predictable.

The assumption concerns complete prefixes. A prefix is the sequence of choices from the start up to a given time. Fix the primitive choices and their independent reference law P . We allow any tape law ν satisfying

$$\nu[w] \leq DP[w]^s \quad \text{for every finite prefix } w, \quad 0 < s \leq 1, \quad D \geq 1.$$

Our source assumption controls the probability of every complete prefix, rather than the next choice conditional on the past. With a uniform q -symbol reference, the first t symbols have min-entropy at least $st \log_2 q - \log_2 D$. Min-entropy is the negative base-two logarithm of the largest outcome probability, a standard measure of weak randomness [CG88]. The guarantee applies at every length, but remains cumulative: randomness accumulated earlier can permit later choices to be certain. Resetting the algorithm does not renew the guarantee. Section 2 gives further examples and explains the scope for generators with a finite seed.

Questions and approach. We ask what independent-source running-time summaries miss, how much cumulative randomness guarantees termination, and how the answers can guide the algorithm’s choices. A *prefix cover* is a collection of finite choice sequences such that every unsuccessful execution begins with one of them. The analysis can stop at an early prefix on one branch and follow another branch further. The cost is $\sum_w P[w]^s$. Classical tree max-flow/min-cut converts the minimum cover cost into the exact worst-case source probability. In the natural prefix metric, the same minimum is

Hausdorff content. Dimension identifies a threshold for nontermination; content gives quantitative bounds at a specified source strength. We obtain the following algorithmic consequences.

An optimal retry order that running-time laws cannot identify. Suppose tasks have fixed success sets, and attempting one task does not change another. For a uniform reference source, the optimal rule is to finish a task with the fewest successful symbols first. For a general reference distribution π , replace the number of successful symbols by their powered mass $w_i = \sum_{a \in B_i} \pi(a)^s$, where B_i is task i 's success set. Always attempting an unfinished task of minimum w_i minimizes worst-case failure at every deadline and every D , against all deterministic history-dependent selectors. Yet all selectors have the same independent-source running-time law; even the sums of powered prefix probabilities agree at every depth (Theorem 3.1). The opening exponential separation is given by Theorem 3.2.

A sharp SAT threshold and matching delay. Suppose one assignment makes at least r literals true in every width- k clause, where $k/2 < r < k$. Under independent uniform choices, the walk takes $O_{k,r}(h)$ expected flips from distance h . Under the prefix constraint, the sharp source threshold over this promise class is $s_* = \log_k(2\sqrt{r(k-r)})$. At and above it,

$$\mathbb{E}_\nu \tau = O_{k,r}(\min\{L^3, L/(s - s_*)\}), \quad L = h + \log D + 1, \quad L/0 = \infty.$$

The upper bound allows arbitrary clause overlap and history-dependent clause selection. On suitable formulas and unsatisfying initial assignments, one source forces the matching delay with probability one for every selector. Thus, at the threshold and fixed D , a walk with linear independent-source expectation can be forced to take cubic time (Theorems 4.1 and 4.3). A fixed margin above the threshold restores linear dependence on L . Below s_* , nontermination is possible even with $D = 1$ (Theorem 4.2); it need not be caused by a periodic tape (Corollary B.2).

An exact criterion for restarting on the same tape. A restart changes the state but supplies no fresh randomness. Nevertheless, for a fixed algorithm, restarting can give exponential tails exactly when its nontermination event has a prefix cover of cost below one. The same condition characterizes whether any restart rule can terminate almost surely against every source with $D = 1$ (Theorem 5.1). For explicit finite-state processes, two linear programs compute a stationary selector minimizing worst-case nontermination and a restart rule whenever robust termination is possible (Theorem 5.2). For SAT, an observable rule resets on leaving a Hamming ball around the initial assignment; it needs a distance bound but not the satisfying assignment. Replay removes the supplied bound (Theorem 4.4 and Appendix B.6).

Correct output as well as termination in noisy search. Under a sufficient prefix guarantee on comparison errors, we give predecessor search with error probability at most η and $O_p((h + \log(D/\eta) + 1)^3)$ expected comparisons, for every fixed reference correctness probability $1/2 < p < 1$. Here h is the correct leaf's depth. The algorithm needs neither h nor the tree height. Longer verification chains at deeper leaves place all wrong outputs beyond one common distance barrier, so a single cover controls the total error without a union bound over leaves. A fixed margin above this sufficient threshold gives linear dependence on the same quantity (Theorem 6.1).

Section 2 gives the model and cover identity; Sections 3–6 develop the applications. Related work and further questions follow in Section 7. All proofs are linked from the contents.

2 Prefix covers and source robustness

Fix a finite alphabet $\mathcal{A}$ with at least two symbols and a full-support distribution π . Let $P = \pi^{\mathbb{N}}$. A random tape is a sequence in $\mathcal{A}^{\mathbb{N}}$; $[w]$ is the set of tapes beginning with a finite word w (a cylinder), and $P[w] = \prod_i \pi(w_i)$ is its reference probability. The symbols represent the actual primitive choices consumed by the algorithm. Their alphabet and interpretation are fixed: grouping or recoding them changes the source assumption.

For $0 < s \leq 1$ and $D \geq 1$, define

$$\mathcal{Q}_{s,D} = \{\nu : \nu[w] \leq DP[w]^s \text{ for every finite } w\}. \quad (1)$$

Here ν ranges over probability laws on whole tapes, and one constant D applies at all lengths. No randomness is reserved for the algorithm outside this tape. Larger s strengthens the guarantee, whereas larger D weakens it. At $s = D = 1$, summing the bounds over every level shows that $\nu = P$. For the uniform law on q symbols, this is exactly the requirement that the first t symbols have min-entropy at least $st \log_2 q - \log_2 D$. A conditional bound q^{-s} on the probability of each next symbol implies membership in $\mathcal{Q}_{s,1}$; the converse is false, as the ternary stream in the introduction shows. Under a nonuniform reference, (1) is a relative probability bound rather than a fixed-rate entropy bound. Conditioning on a positive-probability prefix w gives $\nu[v \mid w] \leq D_w P[v]^s$, where $D_w = DP[w]^s / \nu[w]$. This constant can be much larger than D ; there is no uniform reset of the original guarantee after a history.

Examples and scope. The introductory stream lies strictly above the SAT threshold, so the unmodified walk already has an exponential tail; the restart separations use different sources. Replacing its repeated block by m fixed symbols preserves the prefix bound and makes the coordinates mutually independent. A lack of fresh randomness can therefore arise with or without dependence. Independent bits with bias p satisfy the fair-bit version whenever $\max\{p, 1 - p\} \leq 2^{-s}$.

The bound is information-theoretic, not a test for pseudorandomness. A generator with at most 2^b possible seeds has an output prefix of probability at least 2^{-b} at every length. To satisfy our condition through time T under a uniform q -symbol reference, it must therefore have $b + \log_2 D \geq sT \log_2 q$. A fixed finite seed cannot satisfy the condition forever. This limits which sources the theorem covers; it is not a lower bound on the randomness needed by every SAT solver. Finite-time conclusions require the prefix inequalities only through the deadline in question.

A selector is deterministic and *nonanticipating*: it uses the observed history but no future symbols. Call a history *live* if the algorithm has not terminated. Let τ count updates before termination. In local search, an update attempts to resolve a violated constraint; it need not decrease the total number of violations. Write $\mathcal{N} = \{\tau = \infty\}$. For a closed tape event E , define

$$\mathcal{C}_s(E) = \inf_{\mathcal{B}} \sum_{w \in \mathcal{B}} P[w]^s, \quad E \subseteq \bigcup_{w \in \mathcal{B}} [w], \quad (2)$$

where $\mathcal{B}$ ranges over finite prefix-free covers. Prefix-free means that no chosen word extends another, so the cylinders are disjoint. Their lengths may differ. The empty word has cost one. Every cover gives an upper bound by summing the source bounds over its words. The identity below also gives a converse: some admissible source attains the best such bound.

Theorem 2.1 (Classical tree capacity identity). *For every closed E , $0 < s \leq 1$, and $1 \leq D < \infty$,*

$$\max_{\nu \in \mathcal{Q}_{s,D}} \nu(E) = \min\{1, DC_s(E)\}. \quad (3)$$

If E depends on the first T symbols, a minimizing cover uses words of length at most T . For finite-symbol events $E_T \downarrow \mathcal{N}$,

$$\max_{\nu \in \mathcal{Q}_{s,D}} \nu(\mathcal{N}) = \lim_{T \rightarrow \infty} \max_{\nu \in \mathcal{Q}_{s,D}} \nu(E_T). \quad (4)$$

A source is a probability flow down the prefix tree; a cover cuts every branch in E . Tree max-flow/min-cut gives the identity, as in the tree proof of Frostman's lemma [BP17, Section 3.1]. Appendix A includes the completion of a maximizing subflow to a probability law.

Hausdorff content and dimension. Give two distinct tapes distance $d_P(\omega, \eta) = P[\omega \wedge \eta]$, where $\omega \wedge \eta$ is their longest common prefix; identical tapes have distance zero. A cylinder $[w]$ has diameter $P[w]$, so $\mathcal{C}_s(E)$ is precisely its s -dimensional Hausdorff content. For a closed nontermination set of dimension δ , within $0 < s \leq 1$, every $s < \delta$ permits a source supported on that set for some finite D , whereas every $s > \delta$ forces its probability to zero for every finite D . At a specified D , including the endpoint $s = \delta$, the content in (3) supplies the quantitative answer. Corollary A.1 records the precise statement. The geometry belongs to the set of execution tapes of the specified algorithm.

The cover recursion. For a deadline T , give each surviving depth- T word value one and each terminated word value zero. Work backwards. At a prefix w , either cover everything below it, at normalized cost one, or cover its children separately. The minimum normalized cost is

$$V(w) = \min \left\{ 1, \sum_{a \in \mathcal{A}} \pi(a)^s V(wa) \right\}. \quad (5)$$

The root value is $\mathcal{C}_s(\{\tau > T\})$. The minimum compares paying for the current prefix with continuing down the tree and paying for its surviving descendants. Different branches can make different choices. A single-depth sum forces every branch to stop at the same time.

For example, with fair bits and $s = 1/2$, consider the event $E = [0] \cup [1000]$. A cover that stops at 0 or 1000 costs $2^{-1/2} + 2^{-2} < 1$. Every cover at a single depth costs at least one: depths 0, 1, 2, 3, 4 cost respectively 1, $\sqrt{2}$, $3/2$, $5/\sqrt{8}$, $9/4$, and refining beyond depth four only increases the cost. The example is small, but already shows why even the best choice of a single depth can lose information.

What a single-depth bound retains. Let $\mathcal{L}_T$ be the minimal prefixes completing T updates, including a possible final update that terminates the algorithm. Put

$$Z_T(s) = \sum_{w \in \mathcal{L}_T} P[w]^s. \quad (6)$$

Then $Z_T(1) = P\{\tau \geq T\}$ and $\mathbb{P}_\nu\{\tau \geq T\} \leq DZ_T(s)$. With one symbol per update, let $\tilde{Z}_T(s)$ instead sum over the length- T words still live after the update. For a uniform q -symbol source,

$$Z_T(s) = q^{(1-s)T} P\{\tau \geq T\}. \quad (7)$$

These sums count each level separately. A prefix cover can combine levels and therefore uses how the surviving prefixes extend one another.

3 An optimal order invisible to stopping-time analysis

Suppose attempting one task does not change the others. Under independent randomness, their total running time has the same distribution in every order. Under a global prefix constraint, the order can determine whether a linear deadline is missed with probability one or exponentially small probability. We first identify an optimal order for arbitrary heterogeneous tasks, then quantify this distinction on two task types.

There are n unfinished tasks and a finite symbol alphabet $\mathcal{A}$. Task i has a nonempty success set $B_i \subseteq \mathcal{A}$. An attempt chooses an unfinished task and reads the next symbol. The task completes if the symbol lies in B_i ; otherwise it remains unfinished. Completed tasks are never selected again, and updating one task does not change another. The success sets may overlap. A deterministic nonanticipating selector may use the entire observed history. Let τ count attempts until all tasks complete. For $P = \pi^{\mathbb{N}}$ with full-support π , put

$$W = \sum_{a \in \mathcal{A}} \pi(a)^s, \quad w_i = \sum_{a \in B_i} \pi(a)^s.$$

Theorem 3.1 (Optimal priority under a prefix constraint). *For every $0 < s \leq 1$, always attempting an unfinished task with smallest w_i minimizes $\max_{\nu \in \mathcal{Q}_{s,D}} \mathbb{P}_{\nu}\{\tau > T\}$ over all deterministic nonanticipating selectors, simultaneously for every $T \geq 0$ and $D \geq 1$. If $w_1 \leq \dots \leq w_n$, set $v_t(n+1) = 0$, $v_0(j) = 1$ for $1 \leq j \leq n$, and*

$$v_{t+1}(j) = \min\{1, (W - w_j)v_t(j) + w_j v_t(j+1)\}.$$

The exact optimum is $\min\{1, Dv_T(1)\}$. Once the masses are computed and sorted, this takes $O(nT)$ arithmetic operations and $O(n)$ memory. Nevertheless, every selector has the same stopping-time law under P and the same single-depth prefix power sums for every positive power.

For a uniform reference source, the rule is to finish a task with the fewest successful symbols first. Although every order is equivalent under independent randomness, this order is optimal over the whole source class at every deadline. For a nonuniform reference, w_i is a cover weight: all symbol weights sum to W , which may exceed one. With singleton success sets, the rule still orders tasks by increasing ordinary success probability; in general the order can depend on s . Appendix D.1 proves a weighted exchange inequality compatible with the truncation at one. This is the step that rules out any advantage from history-dependent switching and reduces a dynamic program on 2^n subsets to n suffixes.

A quantitative separation. Take n tasks of type A , successful on 0, and one task of type B , successful on 1. Write Λ_A for completing all A tasks first and Λ_B for completing B first.

Theorem 3.2 (Equal prefix sums, different robustness). *Fix $P(0) = 15/16$, $P(1) = 1/16$, and $\mathcal{Q} = \mathcal{Q}_{1/2,1}$. For every $n \geq 1$,*

$$\max_{\nu \in \mathcal{Q}} \mathbb{P}_{\nu}^{\Lambda_A}\{\tau > 8n\} = 1, \tag{8}$$

$$\left(\frac{15}{16}\right)^{4n} \leq \max_{\nu \in \mathcal{Q}} \mathbb{P}_{\nu}^{\Lambda_B}\{\tau > 8n\} \leq \frac{9}{8} \left(\frac{225}{256}\right)^n < 1. \tag{9}$$

One source $\nu_n \in \mathcal{Q}$ gives survival probability one for Λ_A and at most the upper bound in (9) for Λ_B . All deterministic nonanticipating selectors have identical prefix power sums at every depth and every positive power, and identical stopping-time laws under P . The lower bound holds for every selector; moreover, Λ_B is exactly minimax optimal at every deadline by Theorem 3.1.

Why all power sums agree. For $q > 0$, let $W_q = \sum_a \pi(a)^q$ and $\pi_q(a) = \pi(a)^q/W_q$. The surviving-prefix sum is

$$\tilde{Z}_T^\Lambda(q) = W_q^T \mathbb{P}_{\pi_q^\Lambda}^{\Lambda} \{\tau > T\}.$$

Under this tilted independent law, preassign an independent stream to each task. The total number of attempts is a sum of independent geometric workloads, regardless of the selector. This proves equality for every q , including history-dependent selection. The sums that include a terminating T th attempt agree as well: $Z_0(q) = 1$ and $Z_T(q) = W_q \tilde{Z}_{T-1}(q)$ for $T \geq 1$.

Why the robust tails differ. The bad order postpones the only task that needs a 1. To exploit this, generate zeros with probability $3/4$ until the n th zero, then force zeros through time $8n$ and use P afterwards. The initially random segment makes the deterministic continuation admissible under the prefix constraint. If the A tasks finish by the deadline, task B receives only zeros for the remaining time. For Λ_B , cover first by the prefix 0^{4n} . Otherwise the first 1 completes B by time $4n$, and survival requires fewer than n zeros in the remaining symbols. A generating-function bound covers these remaining paths at their full depth. The resulting mixed-depth cover proves (9); Appendix D.2 checks the source constraint and both bounds.

Thus the cover changes an algorithmic choice that every single-depth power sum leaves unresolved. One direct use is obtaining an accepted proposal for each of several rejection tasks: finish the task with smallest powered acceptance mass first. Those masses must be supplied or efficiently computable. This is a termination guarantee; correlated proposals need not preserve the output distribution of independent rejection sampling.

Sharing a source across components. A global guarantee need not hold on a component's interleaved subsequence. Appendix F gives $N - 1$ graph processes with individual worst-case nontermination probabilities $\exp[-\Theta(N)]$, yet one admissible shared source prevents joint termination under every interleaving. Under the independent reference source, every interleaving has expected total running time $O(N)$. The appendix gives the exact capacity formula.

4 Local SAT search: sharp bounds and restarts

A flip can satisfy one clause while violating several others, so the number of violated clauses is not a reliable measure of progress. The promise below supplies a different measure: distance to an unknown assignment that satisfies a majority of each clause's literals.

Fix integers $k/2 < r < k$. A k -CNF is *strongly r -satisfiable* if some assignment x^* makes at least r literals true in each clause; clauses use k distinct variables. The walk need only find an ordinary satisfying assignment, not this stronger witness. The algorithm chooses any violated clause and flips the variable indexed by the next source symbol. Under the reference source this index is uniform. The clause choice may depend on the complete past. This is an established polynomial-time algorithm for the promise problem [AGH17, Proposition 6.1]. We fix an ordering of each clause's literals and take source symbols in $[k]$, with uniform reference law P . Put

$$\lambda = \sqrt{\frac{r}{k-r}} > 1, \quad s_* = \log_k(2\sqrt{r(k-r)}) \in (0, 1). \quad (10)$$

Theorem 4.1 (Robustness from distance to a solution). *Let Φ be a strongly r -satisfiable k -CNF on n variables, and let h be the Hamming distance of its initial assignment from a strong witness x^* . Put $L = h + \log D + 1$ and $\delta = s - s_*$. There are constants $c, C > 0$, depending only on k, r , such that for every deterministic nonanticipating clause selector, $s \in [s_*, 1]$, $D \geq 1$, $\nu \in \mathcal{Q}_{s,D}$, and integer $T \geq 0$,*

$$\mathbb{P}_\nu\{\tau > T\} \leq \min\{1, CD\lambda^h e^{-cT^{1/3}}, D\lambda^h k^{-\delta T}\}, \quad (11)$$

$$\mathbb{E}_\nu \tau \leq C \min\{L^3, L/\delta\}. \quad (12)$$

We interpret $L/0 = \infty$. Here τ counts variable flips until the formula is satisfied; there is no restriction on clause overlap.

The walk does not know x^* . Under independent randomness its expected flip count is at most $kh/(2r - k)$. The robust bound remains small near a strong solution, regardless of how many other variables the formula contains. The statement counts flips; finding a violated clause takes additional work.

Why different depths help. If a selected clause is violated, at least r of its variables disagree with x^* . Thus at least r labels decrease the distance and the others increase it. The distance process need not be Markov: the number of decreasing labels can vary with the clause and the entire history. For a source exponent s , the two comparison weights are $a_s = rk^{-s}$ and $b_s = (k - r)k^{-s}$. After an exponential change of scale, the total weight per step is $2\sqrt{a_s b_s}$; the threshold s_* is where this equals one. At that point, $a_{s_*} = \lambda/2$ and $b_{s_*} = 1/(2\lambda)$, and dividing the value at distance d by λ^d makes both directional weights $1/2$. The comparison then has the diffusive time scale of a symmetric walk. These are weights in a cover sum, not conditional probabilities of the actual next move.

Choose a barrier $m > h$. Cover a surviving run when it first reaches distance m , or at time T if it has stayed below m . The first part costs at most $(h/m)\lambda^{h-m}$: a distant excursion is exponentially expensive in m . The second costs at most $C\lambda^h e^{-cT/m^2}$: a walk confined to a strip of width m decays on time scale m^2 . Balancing m against T/m^2 gives $m \asymp T^{1/3}$ and the stated tail. The first-hit prefixes have different lengths even though there is only one spatial barrier. Appendix B gives the full argument as a general integer-potential lemma, valid whenever at least r labels decrease a positive potential by one and no label increases it by more than one.

At $s = s_*$, a single-depth argument also proves polynomial termination: reflecting at the global bound n yields $\mathbb{E}_\nu \tau = O_{k,r}(n^2(h + \log D + 1))$. The gain in (11) is its dependence on h in place of n , and the stronger critical tail below. For $h = \Theta(n)$ and fixed D , both expectation bounds have cubic order. The distance barrier uses an unknown witness. Below we replace it by an observable Hamming ball to obtain an efficient restart rule.

Theorem 4.2 (Sharp threshold and a gap from every single depth). *Let Φ_n contain all clauses with r positive and $k - r$ negative literals on distinct variables, with positive literals ordered first, and put $N = n - k + 2 \geq 3$.*

- (i) *For every $0 < s < s_*$, some n , initial assignment, and single source in $\mathcal{Q}_{s,1}$ force nontermination under every nonanticipating clause selector.*
- (ii) *Start with exactly r zero variables. Uniformly in $s \in [s_*, 1]$, $N \geq 3$, and $T \geq 1$, the worst survival probability is*

$$A_{N,T}(s) := \max_{\nu \in \mathcal{Q}_{s,1}} \mathbb{P}_\nu\{\tau > T\} = \exp\left[-\Theta_{k,r}\left(T^{1/3} + (s - s_*)T + T/N^2\right)\right]. \quad (13)$$

At $s = s_*$, for the same event $E_T = \{\tau > T\}$, let $F_t(E_T)$ be the minimum powered mass of a cover using only length- t cylinders. Then

$$\inf_{t \geq 0} F_t(E_T) = \Theta_{k,r}((T+1)^{-3/2}) \quad (1 \leq T \leq N-2). \quad (14)$$

Thus s_* is the sharp uniform threshold for this walk over the promise class. At $s = s_*$ and a deadline proportional to n , variable-depth covers give $e^{-\Theta(n^{1/3})}$, while every single-depth cover of the same event costs at least order $n^{-3/2}$. The symmetric family supplies lower bounds; the upper theorem requires no such symmetry.

For $k = 3, r = 2$, the source condition is particularly simple: $\nu[w] \leq D2^{-3|w|/2}$. The distribution of the first t choices has min-entropy at least $1.5t - \log_2 D$ bits, without assuming positive conditional min-entropy at every step. This promise problem is already tractable: replacing $(a \vee b \vee c)$ by the three pairwise clauses reduces finding a strong assignment to 2-SAT. The result concerns the robustness of the specified random walk, rather than a new tractability threshold.

Nontermination without eventual cycling. The obstruction below s_* need not cycle. Corollary B.2 in Appendix B gives a full-support source with failure probability at least $1 - \varepsilon$ for every selector. For each fixed memoryless selector, the trajectory on failure is almost surely not eventually periodic. Positive probability for every next label therefore does not by itself ensure termination.

The whole transition is sharp. Define

$$\Psi_s(L) = \min\{L^3, L/(s - s_*)\}, \quad L/0 = \infty.$$

Theorem 4.3 (Matching delay across the source threshold). *Fix $k/2 < r < k$. There is a constant $c_{k,r} > 0$ such that for every $s \in [s_*, 1]$, integer $h \geq r$, and $D \geq 1$, some complete formula Φ_n from Theorem 4.2 has a live initial assignment at distance h from its unique strong witness and a single source $\nu \in \mathcal{Q}_{s,D}$ for which, simultaneously for every clause selector,*

$$\tau \geq c_{k,r} \Psi_s(h + \log D + 1) \quad \nu\text{-almost surely.}$$

The number of variables satisfies $n = O_{k,r}(\Psi_s(h + \log D + 1))$. In particular, (12) is sharp, uniformly in initial distance, source slack, and $s \in [s_, 1]$.*

The lower bound strengthens a worst-case expectation statement: one source forces the entire delay, for every selector. At the threshold and fixed D , this delay has order h^3 , although the independent-source expectation is only $O_{k,r}(h)$. Writing $\delta = s - s_* > 0$, the crossover occurs at $\delta L^2 \asymp 1$: the cubic bound L^3 governs smaller L , and L/δ governs larger L . The corresponding time scale is $\delta^{-3/2}$. Thus the theorem quantifies how much the specified walk can slow down when only the cumulative source guarantee remains. For the lower bound in the initial distance, we show that the survival event has capacity exactly one. This gives probability-one delay already with $D = 1$. Near the threshold, we construct a lower bound for the capped recurrence that rises from zero to a flat region of height one. Its boundary at distance m advances by order $\epsilon + m^{-2}$ per step, where $\epsilon = (s - s_*) \log k$. At criticality the speed is of order m^{-2} , so reaching distance j takes cubic time. A positive ϵ eventually gives speed of order ϵ and time j/ϵ . An exponentially tilted sine profile makes this comparison rigorous. Until the boundary reaches the initial distance, the capacity remains exactly one. Tree max-flow then supplies a source supported entirely on executions that survive the required time. For the slack dependence, a separate cover lower bound keeps capacity at least $1/D$ for the required horizon. The two constructions give $L = h + \log D + 1$ in Theorem 4.3. The cube-root-to-linear displacement transition is known in branching processes [Mal19, BLMS24]; Lemma B.3 proves the discrete comparison needed for the admissible-tape construction.

An observable restart rule. The cover proof identifies excursions far from a strong solution as an obstruction to a better tail. Although the solution is unknown, a radius bound lets us cut off these excursions using only the current and initial assignments.

Theorem 4.4 (Restarting on leaving a Hamming ball). *Suppose the formula has a strongly r -satisfying assignment x^* at Hamming distance $h \geq 1$ from the initial assignment x_0 . Let $H \geq h$ be a supplied positive integer. Run the violated-clause walk, and reset to x_0 whenever an unsatisfying assignment leaves the Hamming ball of radius $2H$ around x_0 . Continue with the next unused source symbol. The rule uses H and x_0 , but does not use x^* . There are constants $c, C > 0$, depending only on k, r , such that for every clause selector, $D \geq 1$, $s \in [s_*, 1]$, and $\nu \in \mathcal{Q}_{s,D}$, its stopping time τ_H satisfies*

$$\mathbb{P}_\nu\{\tau_H > T\} \leq \min\{1, CD\lambda^h e^{-cT/H^2}\}, \quad T \in \mathbb{Z}_{\geq 0}. \quad (15)$$

In particular,

$$\mathbb{E}_\nu \tau_H = O_{k,r}\left(H^2(h + \log D + 1)\right).$$

The bound counts source-driven flips. A list of changed coordinates maintains the distance from x_0 in constant additional work per flip. Including resets, the total number of variable updates is at most twice the number of source-driven flips. Work to find a violated clause is separate.

At $s = s_*$, for a failure probability at most ε , the bound gives

$$O_{k,r}\left(H^2(h + \log D + \log(1/\varepsilon) + 1)\right)$$

flips, compared with $O_{k,r}((h + \log D + \log(1/\varepsilon) + 1)^3)$ for the original walk. For $H \asymp h$ and fixed D , both expectation guarantees are cubic; the improvement is the dependence on confidence.

The radius bound H is part of the guarantee: the algorithm needs neither the nearby witness nor its distance h , but it does need a valid upper bound on that distance. Taking $H = n$ is always valid but triggers no resets. A smaller supplied bound gives a tail scale H^2 , independently of the number of variables. No enumeration of assignments or capacity computation is required.

To see why the rule works, let y be an assignment just outside the ball. Then $\text{dist}(y, x^*) \geq 2H + 1 - h > h$. Resetting to x_0 therefore decreases the unknown witness distance. The restarted walk retains the favorable-flip comparison and keeps that distance at most $h + 2H$. A single-depth spectral estimate on this bounded range gives the exponential tail. Thus the benefit of the barrier analysis is an observable change to the algorithm; the changed algorithm then admits a simpler analysis. Appendix B gives the full proof.

For the complete formula family of Theorem 4.2, starting with r zeroes, every favorable first flip satisfies the formula. Restarting after each unsuccessful flip therefore has exact tail

$$\max_{\nu \in \mathcal{Q}_{s_*, D}} \mathbb{P}_\nu\{\tau_{\text{cutoffl}} > T\} = \min\{1, D[(k - r)k^{-s_*}]^T\}.$$

This example shows the benefit of restarting itself. The Hamming-ball theorem applies to arbitrary promised formulas, where a solution may require several flips. Proposition 5.3 separately shows an advantage of adaptive restarts over predetermined cutoffs.

Removing the supplied radius by replay. The supplied radius can be removed by running the original walk alongside copies with geometrically increasing radii. Each copy reads the same archived tape from its beginning, at its own pace. A fixed allocation of work gives an exponential tail with scale $h^2(1 + \log(h + 1))^2$ and preserves the $O_{k,r}(\Psi_s(L))$ expectation bound, where $L = h + \log D + 1$, counting all flips and resets. The construction needs neither a supplied radius nor D , but uses $O(\log(n + 2))$ search states and tape storage. Theorem B.4 in Appendix B.6 gives the precise guarantees and an explicit schedule. No independence between copies is assumed.

5 When restarting restores exponential tails

A restart changes the algorithm's state, but does not replenish the randomness in the tape. A prefix cover nevertheless tells us when to abandon an attempt: restart when the observed word reaches the cover. The criterion below shows that robust almost-sure termination and an exponential tail are possible under exactly the same strict cover inequality.

Fix an initial configuration x and a deterministic algorithm that consumes one source symbol per step. A restart resets the algorithm to x and resumes with the next unused symbol. The policy is deterministic, uses only observed symbols, and may restart at most once between successive steps. Running time counts source-consuming steps; reset costs are separate. Write $\mathcal{N}_x$ for the original nontermination event and $\alpha_s(x) = \mathcal{C}_s(\mathcal{N}_x)$, where $0 < s \leq 1$.

Theorem 5.1 (Exact restart criterion). *The following statements are equivalent:*

- (i) $\alpha_s(x) < 1$.
 - (ii) *Some restart policy and constants $K < \infty$, $0 < r < 1$ satisfy $\Pr_\nu\{\tau_{\text{restart}} > t\} \leq DKr^t$ for every $D \geq 1$, $\nu \in \mathcal{Q}_{s,D}$, and integer $t \geq 0$.*
 - (iii) *Some restart policy terminates almost surely for every $\nu \in \mathcal{Q}_{s,1}$.*
- If $\alpha_s(x) = 1$, every restart policy has nontermination probability one under some source in $\mathcal{Q}_{s,1}$.*

Proof of sufficiency. If $\alpha_s(x) < 1$, continuity of finite-horizon capacities gives a horizon T and a finite prefix-free cover $\mathcal{B}$ of $\{\tau > T\}$, with words of length at most T , whose cost $c = \sum_{b \in \mathcal{B}} P[b]^s$ is less than one. Remove words on which termination has already occurred. Restart on reaching a word in $\mathcal{B}$. Every attempt ends within T steps. The words describing k failed attempts form the prefix-free family $\mathcal{B}^k$, of total cost c^k . Consequently,

$$\Pr_\nu\{\tau_{\text{restart}} > kT\} \leq D \sum_{w \in \mathcal{B}^k} P[w]^s = Dc^k.$$

This uses the source bound on whole execution prefixes; it requires no conditional entropy guarantee after a restart. The converse is proved in Appendix C. $\square$

For a finite explicit state space, one can synthesize the updates as well as the restart rule. Let X be the live states, let F_y be a finite nonempty set of available updates at y , and let $\Phi_f(y, a)$ be the state reached when update f reads symbol a . Set

$$R_f(y, z) = \sum_{a: \Phi_f(y, a) = z} \pi(a)^s, \quad y, z \in X,$$

omitting terminated states. Updates are selected before reading the next symbol; selectors have no additional private randomness.

Theorem 5.2 (Synthesis of robust selection and restarts). *Let V be the unique optimizer of*

$$\max \left\{ \sum_{y \in X} v(y) : 0 \leq v \leq \mathbf{1}, \quad v(y) \leq \sum_{z \in X} R_f(y, z)v(z) \text{ for every } y \in X, f \in F_y \right\}. \quad (16)$$

There is a deterministic stationary selector μ such that, for every initial state x and every $D \geq 1$,

$$\max_{\nu \in \mathcal{Q}_{s,D}} \mathbb{P}_\nu^{\mu,x}\{\tau = \infty\} = \min_{\Lambda} \max_{\nu \in \mathcal{Q}_{s,D}} \mathbb{P}_\nu^{\Lambda,x}\{\tau = \infty\} = \min\{1, DV(x)\},$$

where Λ ranges over deterministic nonanticipating selectors without restarts.

Starting from a fixed reset state x_0 , some deterministic nonanticipating policy, allowed to select updates and restart to x_0 , terminates almost surely for every source in $\mathcal{Q}_{s,1}$ if and only if $V(x_0) < 1$. When this holds, using μ and restarting on entry to $S = \{y : V(y) = 1\}$ gives constants $K < \infty$ and $0 < r < 1$ such that $\mathbb{P}_\nu\{\tau_{\text{restart}} > t\} \leq DKr^t$ for every $D \geq 1$, $\nu \in \mathcal{Q}_{s,D}$, and integer $t \geq 0$. If $V(x_0) = 1$, every such policy has nontermination probability one under some source in $\mathcal{Q}_{s,1}$. For rational powered coefficients $R_f(y, z)$, V , μ , the restart set, and explicit constants for the tail bound are computable in polynomial time in the explicit state-action description.

Affine monotone dynamic programming provides the classical LP and contractive-policy machinery [Ber19]. Here it yields exact guarantees against globally constrained tapes, including an impossibility result for all history-dependent policies. The first linear program determines the optimal values, but choosing an update that satisfies the value equation need not attain them. At a state with $V = 0$, an update that always terminates and one that loops forever can both satisfy that equation. A second linear program resolves these ties. On states with $V < 1$, it chooses among these tied updates and produces a positive potential h with $h = \mathbf{1} + Ah$, where A is the selected powered transition matrix restricted to those states. Thus $Ah < h$, which forces geometric decay. Adjusting the potential also proves decay after transitions into $\{V = 1\}$ are redirected to the reset state. The proof in Appendix C sums powered probabilities of complete prefixes, without assuming fresh randomness after a restart. An implicit state space can be exponentially large. The Hamming-ball rule of Theorem 4.4 avoids this enumeration by using the geometry of the assignments.

Proposition 5.3 (Adaptive versus predetermined restarts). *A three-live-state procedure has reference mean running time $8/3$ under a uniform four-symbol source and the following properties. For every predetermined sequence of positive integer cutoffs, allowing a final infinite cutoff, some source in $\mathcal{Q}_{1/2,1}$ gives nontermination probability at least $1/2$. In contrast, one stationary restart rule satisfies, for every $D \geq 1$ and integer $T \geq 2$,*

$$\max_{\nu \in \mathcal{Q}_{1/2,D}} \Pr_\nu\{\tau_{\text{adaptive}} > T\} = \min\{1, DT2^{1-T}\}.$$

Restart policies based on elapsed time and observations are established. Luby, Sinclair, and Zuckerman study universal schedules for independent runs [LSZ93]; dynamic policies and models of dependence among runs are studied in [KHR⁺02, RHK02]. Here the capacity criterion and the separation concern one global prefix constraint, which supplies no fresh randomness at a reset. The separating example is given in Appendix C.

6 Noisy predecessor search under a prefix constraint

Noisy search must both finish and return the right answer. Under independent errors, repeated checks can provide confidence. Under our source condition, consecutive errors may be predictable from the past, so such checks need an analysis of their complete prefixes. We use verified backtracking and verification chains, which are classical [FRPU94, Section 3]. Our chain lengths depend on leaf depth, giving a guarantee in terms of the correct leaf's depth without knowing it or the tree height. Result-sensitive and sharper bounds under independent noise are established [EGW19, DLU25].

Let $x_1 < \dots < x_n$, $n \geq 1$, be sorted keys and let $x \in \mathbb{R}$ be the query. A primitive query asks whether $x < x_i$. Write $e_t = 1$ when its answer is flipped and $e_t = 0$ otherwise. Fix $1/2 < p < 1$ and assume, for every raw error prefix e ,

$$\nu[e] \leq D(p^{|e|-|e|_1}(1-p)^{|e|_1})^s. \quad (17)$$

Thus the reference law P has independent errors of probability $1 - p$; the actual law need not give any conditional correctness guarantee. Choose the least odd $\ell = \ell(p)$ for which the majority of ℓ independent reference answers is correct with probability greater than $2^{-1/3}$. Repeat each logical comparison ℓ times and take its majority. Let $G \subseteq \{0, 1\}^{3\ell}$ contain the raw error blocks in which all three majorities are correct, and put

$$a_s = \sum_{e \in G} P[e]^s, \quad b_s = \sum_{e \in \{0,1\}^{3\ell} \setminus G} P[e]^s, \quad 2\sqrt{a_{s_c} b_{s_c}} = 1, \quad \lambda = \sqrt{a_{s_c}/b_{s_c}}.$$

The solution $s_c \in (0, 1)$ is unique and $\lambda > 1$. These are sums over raw patterns; we assume no independence between the observed majority answers.

Theorem 6.1 (Verified search with correlated errors). *Take any ordered binary search tree whose leaves represent the predecessor intervals, and let h be the depth of the correct leaf. For every $D \geq 1$, $0 < \eta < 1$, $s \in [s_c, 1]$, and law satisfying (17), the following algorithm halts almost surely and returns the predecessor of x with error probability at most η . Put $L = \max\{1, \lceil \log_\lambda(D/\eta) \rceil\}$ and attach a chain of depth $(u) + L$ edges to each original leaf u . Starting at the root, verify the current interval and backtrack if verification fails; otherwise descend to the indicated child, or advance along the chain. Halt at the first chain endpoint and return its predecessor label. Each move uses exactly 3ℓ primitive queries, as specified in Appendix E. The number Q of primitive queries satisfies*

$$\mathbb{E}_\nu Q \leq C_p \min \left\{ K^3, \frac{K}{s - s_c} \right\}, \quad K = h + \log D + \log(1/\eta) + 1,$$

where $K/0 = \infty$. The algorithm uses p, D, η , but needs neither h nor a bound on the tree height; chains are implemented by counters.

The depth-dependent chains have a specific purpose. Measure distance from the endpoint of the correct leaf's chain. Every block in G moves one step toward it. The chain lengths put every wrong endpoint beyond a common distance barrier: before any incorrect output, the walk must pass a live state whose distance exceeds its starting distance by more than L . One first-hit cover therefore bounds all incorrect outputs together, with no union bound over leaves. The correct endpoint is used only in the proof; the algorithm does not know it.

The exponent s_c is a sufficient threshold: a block with any wrong majority is treated as potentially adverse. The repetition count is constant for fixed p ; it is included in the query bound. For a balanced tree $h = O(\log(n + 1))$. On an unbalanced tree, the guarantee depends on the queried leaf's depth and requires no prior bound on it. The assumption concerns raw error prefixes; error marginals alone do not imply it.

7 Related work and further questions

Weak-source computation asks what algorithms can do with guarantees weaker than independence [CG88, DMOZ23]. Doron, Moshkovitz, Oh, and Zuckerman [DMOZ25, Definition 2] study *unpredictable sources*: averaged over positions and draws of the source, the realized next symbol has large conditional probability only rarely. Their model permits predictable steps and includes arbitrary finite sources of sufficiently high min-entropy rate, for suitable block parameters. They analyze entropy accumulation on lossless expanders and construct online condensers. Our source class overlaps with this setting. For a uniform binary reference it requires $H_\infty(X_{1:t}) \geq st - \log_2 D$ at every length t , so finite truncations have high min-entropy rate when s is close to one and t is sufficiently large. A bound at one terminal horizon need not constrain short prefixes this way. Here

we determine exact survival probabilities and use them to choose retry orders and restarts on the same tape. Related work on Markov-chain robustness controls changes to individual transition probabilities [Zuc26].

The source-capacity identity is classical tree max-flow/min-cut [BP17]; its dimension interpretation draws on connections among symbolic spaces, entropy, and Hausdorff dimension [Sta93, MW88, Sim15]. Lutz [Lut11] studies constructive and finite-state versions of Billingsley dimension, which measure dimension relative to a reference law. Our covers and source laws carry no computability restriction. Achlioptas, Iliopoulos, and Vlassis [AIV17, Section 4] cut each trajectory when its probability crosses a threshold, using prefixes of different lengths in their entropy-compression analysis. The capped recurrence has the familiar multiplicative optimal-stopping form [JPS21], and the finite-state synthesis uses affine monotone dynamic programming [Ber19]. Here the capacity identity (3) gives exact source probabilities and guides updates and restarts.

Two perspectives from learning theory motivated this formulation. Li and Xu [LX26, Section 3.3] combine local priors on reference subspaces with a prior over the space of subspaces, producing a data-independent prior for pointwise generalization bounds. This retains the hierarchy of local descriptions. Xu [Xu26, Definition 2.2 and Proposition 2.3] formalizes history summaries that preserve the actions, observation laws, costs, and updates needed for a Bellman recursion. Here the analogous question is which execution summaries suffice for control. Separate summaries at every depth can lose the way prefixes extend one another. The connection to both works is conceptual; our proofs use the prefix-source constraint and tree capacity.

The promise-SAT walk and its distance analysis are due to Austrin, Guruswami, and Håstad [AGH17, Proposition 6.1]. Search within Hamming balls is also established [DHW03]; here repeated resets return to the same center under one cumulative tape constraint. The noisy-search precedents are discussed in Section 6.

The cubic scale and its crossover have precedents in probability, statistical physics, and algorithmic search. The asymptotic analysis of Derrida and Simon [DS07] predicts critical survival of order $\exp[-\Theta(t^{1/3})]$ from fixed positive distance for branching with an absorbing wall, with a near-critical crossover governed by approximate boundary motion of the form $m'(t) = \epsilon + c/m(t)^2$. Mallein’s consistent-displacement theorem [Mal19, Theorem 1.1] implies the critical probability-one plateau used here, as explained after Lemma B.3. Berestycki, Liu, Mallein, and Schweinsberg [BLMS24] prove a related transition between cube-root and linear displacement at time of order $\epsilon^{-3/2}$ for branching Brownian motion. Killed-branching estimates also enter computational complexity: Pemantle [Pem09] uses them to bound the cost of finding nearly optimal paths in random trees. Our contribution is the sharp source-robust characterization of the specified SAT walk. The upper bound allows arbitrary clause overlap and history-dependent selection; the matching construction gives one admissible source forcing the delay with probability one for every selector. The proof establishes exact saturation of the discrete capped recurrence and uses the capacity identity to construct the source.

The broader question is how prefix covers can guide adaptive computation when randomness is shared across decisions. Can compact representations replace explicit enumeration of the state space? For reinforcement learning, what additional assumptions permit learning a policy that reaches a target state reliably when the reference transition model is unknown and the random inputs satisfy a cumulative prefix constraint? The present synthesis theorem assumes the model is given. For search with externally verified candidates, can covers guide branching and restarting when proposal distributions adapt to the observed history? This would require extending the fixed reference-source analysis. Within the present model, we ask for the sharp noisy-search threshold, a single SAT walk matching the replay guarantee without tape storage, and natural algorithm families for which any fixed number of cover depths gives an asymptotically weaker bound.

A Prefix covers and worst-case sources

Throughout this appendix, $P = \pi^{\mathbb{N}}$ is the full-support finite source from Section 2. The source class $\mathcal{Q}_{s,D}$ imposes its bound on every prefix of the full tape. Symbols after termination may be continued with the reference law.

Proof of Theorem 2.1. Every prefix cover gives $\nu(E) \leq D \sum_{w \in \mathcal{B}} P[w]^s$, which proves the upper bound. Suppose first that E is determined by T symbols. In the full source tree through depth T , give the root capacity $c(\emptyset) = 1$ and every other node w capacity $c(w) = DP[w]^s$. Route flow from the root only into the depth- T words whose cylinders lie in E . The tree max-flow/min-cut theorem gives a largest admissible flow f of value $\min\{1, DC_s(E)\}$. Indeed, cuts are exactly prefix covers, with root cost 1; a cover of a whole depth- T cylinder by longer words cannot lower its cost, since for nonnegative x_i and $s \leq 1$, $\sum_i x_i^s \geq (\sum_i x_i)^s$.

It remains to extend f to a probability law on all depth- T words. Extend f by zero on branches outside its support. For every internal node, including the root,

$$\sum_a c(wa) \geq c(w).$$

For nonroot nodes this follows from $\sum_a \pi(a)^s \geq 1$; at the root it also uses $D \geq 1$. Since f conserves flow, the residual capacities $r(w) = c(w) - f(w)$ satisfy

$$\sum_a r(wa) \geq r(w).$$

Start an additional flow g of value $1 - f(\emptyset)$ at the root. Whenever its incoming mass at w is at most $r(w)$, distribute that mass among its children without exceeding their residual capacities. The displayed inequality makes this possible at every step. Then $f + g$ has total mass one and respects all node capacities. It cannot give E more than the maximal value of f , because its restriction to successful leaves would itself be a larger admissible flow. Thus this probability law attains the upper bound. Continue it beyond depth T using the reference conditional law. For any suffix v ,

$$\nu[wv] = \nu[w]P[v] \leq DP[w]^s P[v]^s = DP[wv]^s,$$

so it belongs to $\mathcal{Q}_{s,D}$ on the infinite tape space.

For a closed E , let E_n be the union of the length- n cylinders that intersect E . These are decreasing closed-and-open sets with intersection E . Their contents decrease to $\mathcal{C}_s(E)$: monotonicity gives one inequality, while any finite prefix cover of E covers all sufficiently large E_n , by compactness. The class $\mathcal{Q}_{s,D}$ is nonempty and weakly compact. Indeed, $P \in \mathcal{Q}_{s,D}$, the tape space is compact, and each cylinder constraint is closed under weak convergence. Choose maximizers ν_n for E_n and a weakly convergent subsequence with limit $\nu_* \in \mathcal{Q}_{s,D}$. For fixed m and $n \geq m$,

$$\nu_n(E_m) \geq \nu_n(E_n) = \min\{1, DC_s(E_n)\}.$$

Pass to the subsequential limit, using that E_m is closed and open, and then let $m \rightarrow \infty$. Continuity from above yields $\nu_*(E) \geq \min\{1, DC_s(E)\}$, proving attainment and (3). For any decreasing sequence of finite-symbol events E_T with intersection $\mathcal{N}$, the same compactness argument shows $\mathcal{C}_s(E_T) \downarrow \mathcal{C}_s(\mathcal{N})$ and proves (4). $\square$

Corollary A.1 (Dimension as a source threshold). *Let $\mathcal{N}$ be closed and put $\delta = \dim_H^P \mathcal{N}$. For $0 < s < \delta$, some finite D and some $\nu \in \mathcal{Q}_{s,D}$ satisfy $\nu(\mathcal{N}) = 1$. For $\delta < s \leq 1$, every such law gives $\nu(\mathcal{N}) = 0$. For every $0 < s \leq 1$, the least domination constant permitting a law supported on $\mathcal{N}$ is*

$$\inf\{D \geq 1 : \exists \nu \in \mathcal{Q}_{s,D}, \nu(\mathcal{N}) = 1\} = \frac{1}{\mathcal{C}_s(\mathcal{N})}, \quad (18)$$

where $1/0 = \infty$. If $P(\mathcal{N}) = 0$, then

$$\delta = \inf\{s \in (0, 1] : \nu(\mathcal{N}) = 0 \text{ for every finite } D \geq 1 \text{ and every } \nu \in \mathcal{Q}_{s,D}\}. \quad (19)$$

At $s = \delta > 0$, there exists a law with finite domination constant supported on $\mathcal{N}$ exactly when $\mathcal{C}_\delta(\mathcal{N}) > 0$; dimension alone does not settle this endpoint.

Proof of Corollary A.1. In the metric $d_P(\omega, \eta) = P[\omega \wedge \eta]$, each cylinder $[w]$ has diameter $P[w]$. Any nonsingleton set is contained in a cylinder with the same diameter; singleton sets can be covered by cylinders of arbitrarily small diameter. Thus the infimum over countable cylinder covers equals the Hausdorff content. For compact $\mathcal{N}$, every such cover has a finite subcover, so this infimum is $\mathcal{C}_s(\mathcal{N})$. Its zero threshold is therefore $\dim_H^P \mathcal{N}$: the content is positive for $s < \delta$ and zero for $s > \delta$. Theorem 2.1 proves (18) and the strict-threshold claims. If $P(\mathcal{N}) = 0$, then $\mathcal{C}_1(\mathcal{N}) = P(\mathcal{N}) = 0$, so the set in (19) is nonempty and has infimum δ . The same exact identity gives the stated critical criterion. $\square$

A.1 Controlled prefix capacities

For $0 < s \leq 1$, take a finite state space, a common finite symbol alphabet with full-support probabilities $\pi(a)$, and a finite nonempty set F_x of available updates at each live state. Write $P = \pi^{\mathbb{N}}$ and let $\Phi_f(x, a)$ be the state obtained when update f uses symbol a at state x . Set $V_0(x) = 1$ at live states and $V_t(x) = 0$ at terminated states. At live states define

$$V_{t+1}(x) = \min \left\{ 1, \min_{f \in F_x} \sum_a \pi(a)^s V_t(\Phi_f(x, a)) \right\}. \quad (20)$$

Proposition A.2 (Optimal deterministic selection). *For every $D \geq 1$ and finite horizon T ,*

$$\min_{\Lambda} \max_{\nu \in \mathcal{Q}_{s,D}} \mathbb{P}_{\nu}^{\Lambda} \{\tau > T\} = \min\{1, DV_T(x_0)\}, \quad (21)$$

where Λ ranges over deterministic nonanticipating selectors. One selector depending only on current state and remaining time is optimal simultaneously for every $D \geq 1$ and every initial state.

This recursion evaluates a single globally constrained source; it does not impose independent conditional constraints after successive symbols.

Proof of Proposition A.2. Fix a finite horizon. At a live root, a prefix cover either cuts the root at cost one or covers the children after the chosen update. For a specified first update f , the selector can choose its continuation independently after each observed symbol a . The optimal continuation cost is $V_t(\Phi_f(x, a))$, multiplied by the edge cost $\pi(a)^s$. Minimizing over f and comparing with the root cut gives (20) by induction. At the horizon, the surviving event costs one at a live state and zero at a terminated state, exactly the stated boundary values. Each minimum is attained, and an attaining choice can depend only on current state and remaining time. These choices depend on s

but not on D , and the same state-based rule is available from every initial state. Theorem 2.1 now turns the optimal cover cost into (21), since $c \mapsto \min\{1, Dc\}$ is nondecreasing.

This proof optimizes one globally constrained source. It does not reset the domination constant after a prefix or give the source independent conditional choices at each step. The selector has no additional private randomness; allowing trusted private coins would define a different optimization problem. $\square$

B Proofs for the promise-SAT walk

B.1 A barrier bound from an integer potential

The proof has two parts. A harmonic function pays for paths when they first reach a distant barrier. A spectral estimate pays for paths that remain below it. Both bounds are on prefix weights; neither treats the actual source as a Markov chain.

The upper bound uses only a property of the potential after each possible source symbol. In particular, it needs neither a finite state space nor a Markov transition law for the potential.

Lemma B.1 (Integer-potential barrier). *Fix $k/2 < r < k$ and the parameters λ, s_* in (10). Consider a deterministic process driven by symbols in $[k]$, with uniform product reference law P and stopping time τ . Assign to every live history a positive integer potential d , initially $h \geq 1$, and give every terminated history potential zero. At each live history, assume that at least r of the k next symbols either terminate or give a live potential at most $d - 1$, and that every other symbol either terminates or gives a live potential at most $d + 1$. For every integer $m > h$, every $T \geq 0$, every $D \geq 1$, and every $\nu \in \mathcal{Q}_{s_*, D}$,*

$$\mathbb{P}_\nu\{\tau > T\} \leq \min \left\{ 1, D \left[\frac{h}{m} \lambda^{h-m} + \frac{\lambda^h}{\sqrt{\lambda^2 - 1}} \exp \left(-\frac{\pi^2 T}{2(2m-1)^2} \right) \right] \right\}. \quad (22)$$

Consequently the critical tail and cubic expectation bounds in Theorem 4.1 hold.

Proof. Put

$$a = rk^{-s_*} = \lambda/2, \quad b = (k-r)k^{-s_*} = 1/(2\lambda).$$

If f is nonnegative and increasing, with $f(0) = 0$, the sum of f over the live children of a history of potential d , multiplied by the powered edge weight k^{-s_*} , is at most

$$af(d-1) + bf(d+1). \quad (23)$$

Indeed, assign r of the decreasing symbols to the first term. Each other live child has potential at most $d + 1$; terminated children contribute zero. This argument applies separately after every history.

Cover $\{\tau > T\}$ by prefixes that first reach potential m while still live, up to time T , and live length- T prefixes that never reach m . Remove any length- T prefix already in the first set. Because upward increments are at most one, the first set meets every path that leaves $\{1, \dots, m-1\}$. The resulting cover is prefix-free.

For first-hit prefixes use

$$H_m(d) = \frac{d}{m} \lambda^{d-m} \quad (0 \leq d \leq m).$$

This function is increasing, $H_m(0) = 0$, $H_m(m) = 1$, and

$$aH_m(d-1) + bH_m(d+1) = H_m(d) \quad (1 \leq d < m).$$

Start with weight $H_m(h)$ at the root. Expanding a live node below m cannot increase the sum of its powered prefix weight times H_m , by (23). Leave first-hit nodes unexpanded, give terminated nodes value zero, and continue to depth T . Every first-hit node has value one; the remaining values are nonnegative. Finite-tree induction therefore bounds the powered mass of first-hit prefixes by $H_m(h)$.

For paths remaining below m , put $L = m - 1$ and define a reflected comparison matrix $R^{(m)}$ on $\{1, \dots, L\}$ by

$$(R^{(m)}f)(d) = af(d-1) + bf(d+1) \quad (d < L), \quad (R^{(m)}f)(L) = af(L-1) + bf(L),$$

where $f(0) = 0$. This matrix preserves nonnegative increasing vectors: consecutive differences in the interior are nonnegative linear combinations of consecutive differences of f , and the final difference is $a(f(L-1) - f(L-2)) \geq 0$ when $L \geq 2$. The claim is immediate for $L = 1$. Thus $F_t = (R^{(m)})^t \mathbf{1}$ is nonnegative and increasing. Induction on the remaining horizon bounds the powered mass of length- T live paths that stay below m by $F_T(h)$. For a history with potential below L , apply (23) to F_t . At potential L , an upward exit is discarded, each of r decreasing symbols contributes at most $F_t(L-1)$, and any other retained child contributes at most $F_t(L)$. This gives the reflected final row.

Let $B = \text{diag}(\lambda, \dots, \lambda^L)$. The symmetric matrix $B^{-1}R^{(m)}B$ has off-diagonal entries $1/2$, last diagonal entry $b < 1/2$, and other diagonal entries zero. Increasing the last diagonal entry to $1/2$ gives a nonnegative symmetric matrix with largest eigenvalue $\cos(\pi/(2L+1))$. To see the latter, the positive vector $v_j = \sin(j\pi/(2L+1))$ satisfies its eigenvector equation, including the reflected final row. A symmetric nonnegative matrix has Euclidean operator norm equal to its spectral radius. Perron–Frobenius monotonicity consequently gives

$$\|B^{-1}R^{(m)}B\|_2 \leq \cos \frac{\pi}{2m-1}.$$

Since $\|(\lambda^{-j})_{j=1}^L\|_2 \leq 1/\sqrt{\lambda^2-1}$,

$$F_T(h) \leq \frac{\lambda^h}{\sqrt{\lambda^2-1}} \left(\cos \frac{\pi}{2m-1} \right)^T \leq \frac{\lambda^h}{\sqrt{\lambda^2-1}} \exp\left(-\frac{\pi^2 T}{2(2m-1)^2}\right).$$

The source constraint bounds the probability of each cover cylinder by D times its powered reference mass. Summing the two parts proves (22).

When $T^{1/3} > h$, choose an integer $m > h$ within two of $T^{1/3}$. Both terms are at most $C\lambda^h e^{-cT^{1/3}}$. When $T^{1/3} \leq h+1$, the same capped bound follows from probability at most one, after enlarging C and decreasing c . Finally, summing $\min\{1, CD\lambda^h e^{-cT^{1/3}}\}$ over integer $T \geq 0$ gives $C'(h + \log D + 1)^3$. For example, split the sum at a sufficiently large constant times $(h + \log D + 1)^3$; beyond that point its summands are bounded by $e^{-cT^{1/3}/2}$. $\square$

Proof of Theorem 4.1. Let d be the Hamming distance to x^* . Whenever a clause is violated, every literal it satisfies under x^* belongs to a coordinate on which the current assignment and x^* disagree. At least r flips therefore decrease d by one; all others increase it by one. Assign potential zero on termination. Every live assignment has positive potential. The barrier lemma and $\mathcal{Q}_{s,D} \subseteq \mathcal{Q}_{s_*,D}$ give the critical tail and the cubic expectation bound. An initially satisfying assignment has $\tau = 0$.

For the additional bound above the threshold, write $\delta = s - s_*$ and sum over live length- t histories:

$$M_t = \sum_{\substack{|w|=t \\ \tau(w) > t}} k^{-st} \lambda^{d(w)}.$$

At each live history the sum of the children's potential weights is at most

$$k^{-s}(r\lambda^{d-1} + (k-r)\lambda^{d+1}) = k^{-\delta}\lambda^d.$$

Terminated children contribute zero. Thus $M_T \leq \lambda^h k^{-\delta T}$ by finite-tree induction. Every live leaf has $\lambda^{d(w)} \geq 1$, so the prefix constraint gives $\mathbb{P}_\nu\{\tau > T\} \leq DM_T$. For $\delta > 0$, summing the capped exponential tail gives $O_{k,r}(L/\delta)$, where $L = h + \log D + 1$. Combining this with the cubic bound proves (12). This weighted-tree argument also applies to the general potential in Lemma B.1; it uses no conditional probability assumption on the source. Under P , the expected decrease in distance at a live history is at least $(2r - k)/k$, giving $\mathbb{E}_P \tau \leq kh/(2r - k)$ by additive drift. $\square$

The single-depth baseline. For SAT, live distances are at most n . Use the reflected matrix above on $\{1, \dots, n\}$. At distance n , every actual flip decreases distance, so its powered contribution on an increasing vector f is at most $(a + b)f(n - 1) \leq af(n - 1) + bf(n)$. The same induction and spectral estimate therefore bound the single-depth live-prefix sum by

$$\tilde{Z}_T(s_*) := \sum_{w \in [k]^T: \tau(w) > T} k^{-s_* T} \leq \frac{\lambda^h}{\sqrt{\lambda^2 - 1}} \exp\left(-\frac{\pi^2 T}{2(2n + 1)^2}\right). \quad (24)$$

Applying the source bound at depth T and summing the capped tail proves $\mathbb{E}_\nu \tau = O_{k,r}(n^2(h + \log D + 1))$.

B.2 A family with a sharp source threshold

For $n \geq k + 1$, let Φ_n contain every clause

$$\left(\bigvee_{i \in I} x_i\right) \vee \left(\bigvee_{j \in J} \neg x_j\right), \quad I \cap J = \emptyset, \quad |I| = r, \quad |J| = k - r. \quad (25)$$

There are $\binom{n}{k} \binom{k}{r} = O_k(n^k)$ clauses. The all-one assignment strongly r -satisfies the formula. It is the unique such assignment. For an assignment with $z \geq 1$ zeroes and $o = n - z$ ones, if $o \geq k - r$, put $k - r$ ones in the negative positions and at least one zero in the positive positions; the clause has fewer than r true literals. If $o < k - r$, put all the ones in negative positions and fill the remaining positions with zeroes; the clause has $k - r - o < r$ true literals. The required choices exist because $n \geq k$.

Order positive literals first in each clause. If d is the number of zero variables, a clause is violated precisely when its positive variables are zero and its negative variables are one. Hence an assignment is live exactly when $r \leq d \leq n - k + r$. Put $N = n - k + 2$ and $j = d - r + 1$; the live values are $1, \dots, N - 1$. For every selector, symbols $1, \dots, r$ decrease j , and the other symbols increase it. The powered matrix of live transitions is

$$R_s(j, j - 1) = rk^{-s}, \quad R_s(j, j + 1) = (k - r)k^{-s}, \quad (26)$$

with out-of-range entries omitted.

Proof of Theorem 4.2(i). The spectral radius of (26) is

$$\rho_s = 2\sqrt{r(k - r)}k^{-s} \cos(\pi/N).$$

For $s < s_*$, take N sufficiently large that $\rho_s > 1$. A positive right eigenvector is $\varphi(j) = \lambda^j \sin(j\pi/N)$. Start at a maximizing j_0 , and at live value j give label a probability

$$q(a \mid j) = k^{-s} \frac{\varphi(j + \delta(a))}{\rho_s \varphi(j)}, \quad \delta(a) = \begin{cases} -1, & a \leq r, \\ 1, & a > r, \end{cases}$$

where $\varphi(0) = \varphi(N) = 0$. The eigenvector equation normalizes these probabilities, and the boundary labels have probability zero. For every supported word w of length t , telescoping gives

$$\nu[w] = k^{-st} \rho_s^{-t} \frac{\varphi(j_t)}{\varphi(j_0)} \leq k^{-st} = P[w]^s.$$

The resulting source belongs to $\mathcal{Q}_{s,1}$ and never exits the live interval. The path j_t is determined by source labels alone, independently of the clause selector. Thus this one source forces nontermination for all selectors simultaneously. $\square$

Corollary B.2 (Full support and nonperiodic failure). *For every $0 < s < s_*$ and $0 < \varepsilon < 1$, there are a strongly r -satisfiable formula, a fixed initial assignment, and a single source $\nu \in \mathcal{Q}_{s,1}$ such that*

$$\nu\{\tau = \infty\} \geq 1 - \varepsilon$$

for every deterministic nonanticipating clause selector. The source can be generated by a time-homogeneous stochastic machine with finitely many internal states, satisfies $\nu[wa]/\nu[w] > 0$ for every finite word w and every symbol a , and almost surely produces a tape that is not eventually periodic. For each fixed deterministic memoryless clause selector, the assignment trajectory on nontermination is almost surely not eventually periodic.

The conditional probabilities need not have a common positive lower bound. The generator is stochastic, not a deterministic finite-seed PRNG. Nonperiodicity also does not mean that assignments are never revisited: a finite assignment space forces revisits on every infinite run. The statement rules out eventual repetition of a fixed cycle. The proof uses the source from Theorem 4.2(i).

Proof of Corollary B.2. Let $q \in \mathcal{Q}_{s,1}$ be the source in the proof of Theorem 4.2(i), and set $\nu = (1 - \varepsilon)q + \varepsilon P$. Both component laws satisfy the same prefix bound, so ν does too. The q component forces nontermination for all selectors simultaneously, proving the probability bound. The generator for q has the finite internal state $j \in \{1, \dots, N-1\}$ and fixed transition probabilities; an initial choice of mode implements the mixture with P using only finitely many additional states. Furthermore, $\nu[wa] \geq \varepsilon k^{-|w|-1} > 0$.

Every individual tape has mass at most $\lim_{t \rightarrow \infty} k^{-st} = 0$. Since the eventually periodic tapes form a countable set, they have total mass zero. For a memoryless selector, the current assignment determines the selected clause, and the next assignment identifies the flipped coordinate and hence its unique literal index. Thus an eventually periodic assignment trajectory would determine an eventually periodic source tape. $\square$

B.3 Matching tails and the gap from single-depth covers

Proof of Theorem 4.2(ii). Write $R = R_{s_*}$, $a = \lambda/2$, $b = 1/(2\lambda)$, and $\varepsilon = (s - s_*) \log k \geq 0$, so that $R_s = e^{-\varepsilon} R$. The exact distance reduction gives the prefix-capacity recursion

$$C_0 = \mathbf{1}, \quad C_{t+1} = \min\{\mathbf{1}, e^{-\varepsilon} R C_t\},$$

with the minimum taken coordinatewise, and the prefix-cover identity gives $A_{N,T}(s) = C_T(1)$. The critical upper bound in Theorem 4.1, with $h = r$, also applies to $\mathcal{Q}_{s,1} \subseteq \mathcal{Q}_{s_*,1}$: $A_{N,T}(s) \leq C_1 e^{-c_1 T^{1/3}}$. Diagonal similarity transforms R to the symmetric killed simple-walk matrix on $1, \dots, N-1$, whose Euclidean norm is $\cos(\pi/N)$. The full-depth cover therefore gives

$$A_{N,T}(s) \leq e^{-\varepsilon T} e_1^\top R^T \mathbf{1} \leq \frac{\lambda}{\sqrt{\lambda^2 - 1}} e^{-\varepsilon T - \pi^2 T / (2N^2)}.$$

Taking the smaller upper bound gives $C_2 \exp[-c_2(T^{1/3} + \varepsilon T + T/N^2)]$. Since $A_{N,T}(s) \leq A_{N,1}(s) = e^{-\varepsilon} b \leq b < 1$, decreasing the exponent constant absorbs C_2 uniformly for $N \geq 3$, $T \geq 1$, and $s \in [s_*, 1]$.

For the lower bound, retain only paths within $1, \dots, m-1$, where $3 \leq m \leq N$, and define

$$\rho_m = \cos(\pi/m), \quad \varphi_m(j) = \lambda^j \sin(j\pi/m), \quad u(j) = \frac{\varphi_m(j)}{\max_{1 \leq i < m} \varphi_m(i)}.$$

Extend u by zero outside this interval. Then $0 \leq u \leq 1$ and $Ru \geq \rho_m u$. Since $e^{-\varepsilon} \rho_m \leq 1$, induction in the capped recursion gives $C_T \geq e^{-\varepsilon T} \rho_m^T u$; capping at one preserves this lower bound at each step. Furthermore,

$$u(1) \geq \frac{2}{m} \lambda^{2-m}, \quad \rho_m^T \geq e^{-\pi^2 T / m^2} \quad (m \geq 3).$$

Consequently

$$A_{N,T}(s) \geq \frac{2}{m} \lambda^{2-m} e^{-\varepsilon T - \pi^2 T / m^2}.$$

Choose $m = \min\{N, \max\{3, \lceil T^{1/3} \rceil\}\}$. For $T \leq N^3$ the last expression is at least $e^{-C(T^{1/3} + \varepsilon T)}$; for $T > N^3$ it is at least $e^{-C(N + \varepsilon T + T/N^2)}$. Since $N < T/N^2$ in the latter case, this proves (13), with $\varepsilon = (s - s_*) \log k$.

For the single-depth comparison, set $s = s_*$ and write S_N for the symmetric killed simple-walk matrix. Diagonal similarity gives

$$e_1^\top R^t \mathbf{1} = \sum_{j=1}^{N-1} \lambda^{1-j} S_N^t(1, j).$$

The reflection principle on the positive half-line gives, when parities match,

$$S_\infty^t(1, j) = 2^{-t} \left[\binom{t}{(t+j-1)/2} - \binom{t}{(t+j+1)/2} \right] \leq C \frac{j}{(t+1)^{3/2}}.$$

For the inequality, factor the binomial difference as $2j/(t+j+1)$ times its first term and use the central-binomial bound. Since $S_N^t \leq S_\infty^t$ and $\sum_{j \geq 1} j \lambda^{1-j} < \infty$, the single-depth sum is at most $C(t+1)^{-3/2}$.

For even $t = 2\ell \leq N-2$, the Catalan paths from 1 back to 1 stay below N . Their powered mass is $\text{Cat}_\ell(ab)^\ell = \text{Cat}_\ell / 4^\ell \geq c(t+1)^{-3/2}$. For odd $t = 2\ell + 1 \leq N-2$, append one upward step to these paths; this multiplies the mass by b and gives the same order. Thus the single-depth sum is $\Theta_{k,r}((t+1)^{-3/2})$ for every $0 \leq t \leq N-2$.

It remains to compare covers of the same event $E_T = \{\tau > T\}$ at all depths, including depths exceeding T . Every live state has a live neighbor when $N \geq 3$, and hence has an arbitrarily long live continuation. For $t \leq T$, the length- t cylinders meeting E_T are exactly the live words. Therefore $F_t(E_T) = e_1^\top R^t \mathbf{1}$. For $t > T$, all k^{t-T} extensions of each live length- T word belong to E_T , so

$$F_t(E_T) = k^{(1-s_*)(t-T)} e_1^\top R^T \mathbf{1} \geq F_T(E_T).$$

Taking the infimum over all depths proves (14). $\square$

B.4 Sharp dependence on distance and source slack

The capped recurrence retains a region where its value is exactly one. At the critical source exponent its boundary advances on the scale $t^{1/3}$; above the threshold there is an additional linear advance. The exact unit plateau will give a single admissible source that forces a prescribed delay with probability one.

Lemma B.3 (A slowly advancing plateau). *Let $\lambda > 1$, $a = \lambda/2$, and $b = 1/(2\lambda)$. For $\epsilon \geq 0$ define, on the positive integers,*

$$C_0^\epsilon(j) = 1, \quad C_t^\epsilon(0) = 0, \quad C_{t+1}^\epsilon(j) = \min\{1, e^{-\epsilon}(aC_t^\epsilon(j-1) + bC_t^\epsilon(j+1))\}.$$

There are constants $\epsilon_0, c > 0$ and $j_0 \geq 1$, depending only on λ , such that, for $0 \leq \epsilon \leq \epsilon_0$,

$$C_t^\epsilon(j) = 1 \quad \text{if} \quad j \geq j_0, \quad 0 \leq t \leq c \frac{j^3}{1 + \epsilon j^2}.$$

In particular, the permitted horizon has order $\min\{j^3, j/\epsilon\}$, with $j/0 = \infty$.

The critical plateau order has an antecedent in Mallein [Mal19, Theorem 1.1]. Indeed, put $\beta = \log \lambda$ and take a branching random walk with bounded offspring, at least one child per particle, and expected child counts a, b at displacements $+\beta, -\beta$. Its boundary identities are $ae^{-\beta} + be^\beta = 1$ and $a\beta e^{-\beta} - b\beta e^\beta = 0$. The probability of a lineage staying below $j\beta$ through generation t is at most $C_t^0(j)$ by the union bound. Mallein's theorem makes this probability tend to one for $j = \lceil At^{1/3} \rceil \pm 1$ and sufficiently large A ; since $a + b > 1$, the next capped step equals one. The crossover at time of order $\epsilon^{-3/2}$ is also analogous to the near-critical branching Brownian transition studied by Berestycki et al. [BLMS24]. The proof below establishes the needed discrete capped-recursion bound directly; its application here is the construction of sources satisfying all prefix constraints.

Proof. Write $\beta = \log \lambda$. For each sufficiently large real m , let $\theta_m \in (0, \pi/m)$ be the unique solution of

$$m\theta_m + \arctan(\theta_m/\beta) = \pi.$$

Define a function on $[0, \infty)$ by

$$f_m(x) = \begin{cases} e^{\beta(x-m)} \frac{\sin(\theta_m x)}{\sin(\theta_m m)}, & 0 \leq x < m, \\ 1, & x \geq m. \end{cases}$$

This function increases from zero to one: its logarithmic derivative below m is $\beta + \theta_m \cot(\theta_m x) > 0$, and that derivative vanishes at m . Write g_m for the sine expression without truncation and $\rho_m = \cos \theta_m$. Direct calculation gives

$$ag_m(x-1) + bg_m(x+1) = \rho_m g_m(x).$$

For $0 \leq z \leq 1$,

$$g_m(m+z) = e^{\beta z} \left(\cos(\theta_m z) - \frac{\beta}{\theta_m} \sin(\theta_m z) \right) \leq 1;$$

its derivative in z is $-e^{\beta z}(\beta^2/\theta_m + \theta_m) \sin(\theta_m z) \leq 0$. Thus replacing g_m by f_m cannot decrease a neighbor value used at an integer $j < m$. If $F_\epsilon f(j) = \min\{1, e^{-\epsilon}(af(j-1) + bf(j+1))\}$, then

$$F_\epsilon f_m(j) \geq e^{-\epsilon} \rho_m f_m(j) \quad (1 \leq j < m). \quad (27)$$

Choose fixed $z_0 \in (0, 1)$ and $\epsilon_0 > 0$ sufficiently small that

$$e^{-\epsilon_0} \left[a e^{-\beta(z_0+1)} (1 + \beta(z_0 + 1)) + b \right] > 1.$$

Such a choice is possible because the bracket at $z_0 = 0$ equals $(1 + \beta + e^{-\beta})/2 > 1$. For every fixed $z \geq 0$, $f_m(m - z) \rightarrow e^{-\beta z}(1 + \beta z)$. Consequently, for all sufficiently large m and uniformly in $0 \leq \epsilon \leq \epsilon_0$,

$$F_\epsilon f_m(j) = 1 \quad (j \geq m - z_0). \quad (28)$$

Indeed, monotonicity gives $f_m(j - 1) \geq f_m(m - z_0 - 1)$, while $f_m(j + 1) = 1$.

Away from this saturated band, increasing m decreases f_m at a uniform logarithmic rate. Differentiating the equation for θ_m gives

$$-\frac{\theta'_m}{\theta_m} = \frac{1}{m + \beta/(\beta^2 + \theta_m^2)}.$$

For fixed $0 < x < m$ it follows that

$$-\partial_m \log f_m(x) = \frac{\theta_m x \cot(\theta_m x) + \beta m}{m + \beta/(\beta^2 + \theta_m^2)}. \quad (29)$$

Since $y \cot y$ decreases on $(0, \pi)$, the right side for $x \leq m - z_0$ is bounded below by its value at $x = m - z_0$. That value converges to $\beta^2 z_0 / (1 + \beta z_0) > 0$. Hence there are $m_0 \geq 1$ and $\eta > 0$ such that $-\partial_m \log f_m(x) \geq \eta$ whenever $m \geq m_0$ and $0 < x \leq m - z_0$. Enlarge m_0 so that all the preceding comparisons hold and $\rho_m > 0$ for $m \geq m_0$.

For fixed $\epsilon \in [0, \epsilon_0]$, set

$$\Delta_m = \frac{\epsilon - \log \rho_m}{\eta}.$$

For $j < m - z_0$, integrating (29) over $[m, m + \Delta_m]$ gives

$$f_{m+\Delta_m}(j) \leq e^{-\epsilon} \rho_m f_m(j) \leq F_\epsilon f_m(j).$$

For the remaining j , use (28). Thus $F_\epsilon f_m \geq f_{m+\Delta_m}$ pointwise. Since $\theta_m < \pi/m$, there is a constant A such that, uniformly in $m \geq m_0$ and $\epsilon \in [0, \epsilon_0]$,

$$0 < \Delta_m \leq A(\epsilon + m^{-2}).$$

Starting from m_0 , define $m_{t+1} = m_t + \Delta_{m_t}$. Monotonicity of F_ϵ and $C_0^\epsilon \geq f_{m_0}$ give $C_t^\epsilon \geq f_{m_t}$ by induction.

We now bound this discrete sequence. Put $Q = A(\epsilon_0/m_0 + m_0^{-3})$ and $K = A(3 + 3Q + Q^2)$. Because $\Delta_m/m \leq Q$,

$$(m + \Delta_m)^3 - m^3 \leq (3 + 3Q + Q^2)m^2 \Delta_m \leq K(1 + \epsilon m^2).$$

For any $j \geq 2m_0$, suppose there is a first index $u \leq t$ with $m_u > j$. The preceding values are at most j , so summing the last inequality yields

$$j^3 < m_u^3 \leq m_0^3 + Ku(1 + \epsilon j^2) \leq m_0^3 + Kt(1 + \epsilon j^2).$$

This is impossible if $t \leq cj^3/(1 + \epsilon j^2)$, where $c \leq 1/(2K)$, because $m_0^3 \leq j^3/8$. Therefore $m_t \leq j$ throughout that horizon and $C_t^\epsilon(j) \geq f_{m_t}(j) = 1$. Taking $j_0 = \lceil 2m_0 \rceil$ proves the lemma. $\square$

Proof of Theorem 4.3. Write $\delta = s - s_*$, $\epsilon = \delta \log k$, $\ell = \log D$, and $j = h - r + 1 \geq 1$. The source-label sequence determines the distance process of Φ_n independently of the clause selector. For an integer horizon $T \geq 0$, choose $N = j + T + 2$ and $n = N + k - 2$. No path starting at j reaches the upper boundary by time T . Since $R_s = e^{-\epsilon} R_{s_*}$, the finite-interval capacity at (j, T) therefore equals $C_T^\epsilon(j)$. The finite-tree capacity identity, with root capacity one and all nonroot capacities multiplied by D , gives

$$\max_{\nu \in \mathcal{Q}_{s,D}} \Pr_{\nu} \{\tau > T\} = \min\{1, DC_T^\epsilon(j)\}. \quad (30)$$

The maximum is attained. A maximizing finite-depth law extends to an infinite admissible source by appending independent uniform symbols: for $t \geq T$, its prefix bound is $Dk^{-sT}k^{-(t-T)} \leq Dk^{-st}$ because $s \leq 1$.

First suppose $0 \leq \epsilon \leq \epsilon_0$. Lemma B.3 gives $C_T^\epsilon(j) = 1$ whenever $j \geq j_0$ and

$$T \leq c \frac{j^3}{1 + \epsilon j^2}.$$

This supplies a probability-one delay of order $\min\{j^3, j/\epsilon\}$ for large j . To treat slack, monotonicity of the recurrence shows that $C_t^\epsilon(j)$ is increasing in j . The elementary inequality $\min\{1, \alpha x\} \geq \alpha \min\{1, x\}$ for $0 \leq \alpha \leq 1$ also gives, by induction,

$$C_T^\epsilon(1) \geq e^{-\epsilon T} C_T^0(1).$$

The fixed-strip eigenvector lower bound used in Theorem 4.2(ii), with width $\max\{3, \lceil T^{1/3} \rceil\}$, yields

$$C_T^\epsilon(1) \geq \exp[-\epsilon T - B(T+1)^{1/3}]$$

for a constant B depending only on k, r . For all sufficiently large ℓ , choose

$$T = \left\lfloor c' \min\{\ell^3, \ell/\epsilon\} \right\rfloor,$$

again interpreting division by zero as infinity. Choose $c' > 0$ so that $c' \leq 1/4$ and $B(c')^{1/3} \leq 1/4$. Then

$$\epsilon T + B(T+1)^{1/3} \leq c' \ell + B(c')^{1/3} \ell + B \leq \ell$$

for sufficiently large ℓ . Thus $DC_T^\epsilon(j) \geq 1$, and (30) supplies another probability-one delay, now of order $\min\{\ell^3, \ell/\epsilon\}$.

These two horizons combine uniformly. For $f_\epsilon(x) = \min\{x^3, x/\epsilon\}$, monotonicity and $f_\epsilon(ux) \leq u^3 f_\epsilon(x)$ for $u \geq 1$ show that the larger horizon, plus one to cover rounding and bounded parameters, is at least $c_1 f_\epsilon(h + \ell + 1)$. Here $h/r \leq j \leq h$, and $h \geq r$, so the change from j to h costs only a constant depending on r . Because $\epsilon = (s - s_*) \log k$, $f_\epsilon(L) = \Theta_k(\Psi_s(L))$ for $L = h + \ell + 1$. Reducing a fixed constant $c_2 > 0$ therefore gives an admissible source with

$$\Pr_{\nu} \{\tau > \lfloor c_2 \Psi_s(L) \rfloor\} = 1.$$

For bounded parameters the displayed horizon is zero and the initial assignment is live; this also accounts for all floor terms.

It remains to treat $\epsilon \geq \epsilon_0$. The target satisfies $\Psi_s(L) \leq (\log k / \epsilon_0) L$. Distance alone gives survival through $j - 1$ steps under any source, provided the upper boundary is farther than that horizon. For slack, let

$$T_D = \left\lfloor \frac{\ell}{s \log k} \right\rfloor.$$

Use a fixed upward label for the first T_D symbols and independent uniform symbols thereafter. Up to depth T_D , every nonzero cylinder has probability one and satisfies $1 \leq Dk^{-st}$; after that depth, the same extension calculation as above proves all prefix bounds. With $N > j + T_D$ this source forces $\tau > T_D$. The larger of $j - 1$ and T_D , plus one, is at least a constant times $h + \ell + 1$. Reducing c_2 again proves the same probability-one delay throughout the remaining range of s .

Finally take $T = \lfloor c_2 \Psi_s(L) \rfloor$ and $N = j + T + 2$; any source constructed for a longer horizon can be restricted to depth T and extended uniformly. The number of variables is $n = j + T + k = O_{k,r}(\Psi_s(L))$, since $\Psi_s(L) \geq L \geq h$ for $s \in [s_*, 1]$. Moreover $\tau > T$ implies $\tau \geq T + 1 > c_2 \Psi_s(L)$ almost surely, and hence gives the same lower bound on its expectation. All claims hold simultaneously for every selector under the same source because the label-to-distance map is selector independent. $\square$

B.5 An observable restart rule for the promise-SAT walk

Proof of Theorem 4.4. Use the unknown witness distance $d(x) = \text{dist}(x, x^*)$ only in the analysis. Suppose a proposed flip reaches an unsatisfying assignment y outside the ball of radius $2H$ around x_0 . The triangle inequality gives

$$d(y) \geq \text{dist}(y, x_0) - \text{dist}(x_0, x^*) \geq 2H + 1 - h \geq h + 1.$$

Resetting to x_0 therefore strictly decreases this potential. Consequently, after the flip and any reset, each of the at least r favorable symbols still either terminates the walk or decreases d by at least one; every other symbol increases d by at most one. Every live assignment kept by the algorithm lies within distance $2H$ of x_0 , so its potential is at most $L = h + 2H$.

Let $a = \lambda/2$ and $b = 1/(2\lambda)$ as in Lemma B.1. On $\{1, \dots, L\}$ define the reflected comparison matrix

$$(Rf)(d) = af(d-1) + bf(d+1) \quad (d < L), \quad (Rf)(L) = af(L-1) + bf(L),$$

where $f(0) = 0$. Its powers preserve nonnegative increasing vectors. Exactly as in the barrier lemma, induction on the remaining horizon therefore bounds the powered mass of surviving length- T execution words by $e_h^\top R^T \mathbf{1}$. This comparison is valid after every history, including histories containing resets; the clause selector may retain the entire history. The spectral estimate in that lemma gives

$$e_h^\top R^T \mathbf{1} \leq \frac{\lambda^h}{\sqrt{\lambda^2 - 1}} \exp\left(-\frac{\pi^2 T}{2(2L+1)^2}\right).$$

Apply the source constraint to whole length- T execution words and sum. Since $L = h + 2H \leq 3H$, the resulting bound is at most $CD\lambda^h e^{-cT/H^2}$, proving (15) for $s = s_*$; the inclusion $\mathcal{Q}_{s,D} \subseteq \mathcal{Q}_{s_*,D}$ gives the claim for $s \geq s_*$.

Since $L \leq 3H$, summing the capped exponential tail yields $O_{k,r}(H^2(h + \log D + 1))$. More explicitly, for $A > 0$ and $q > 0$, $\sum_{T \geq 0} \min\{1, Ae^{-qT}\}$ is at most $1 + q^{-1}(1 + \max\{0, \log A\})$ by comparison with the integral of the decreasing summand. Here $q^{-1} = O(H^2)$ and $\log A = h \log \lambda + \log D - \frac{1}{2} \log(\lambda^2 - 1)$.

For the implementation claim, keep a list of coordinates on which the current assignment differs from x_0 , together with an inverse index for constant-time insertion and deletion. A flip updates this list in constant time. At a reset, restore and remove every listed coordinate. Each restored coordinate was flipped at least once in that attempt, so the reset uses no more variable updates than the attempt used source-driven flips. Summing over disjoint attempts proves the factor-two bound. The list has size at most $2H + 1$ before a reset. $\square$

B.6 A radius-free portfolio using an archived tape

Theorem B.4 (Unknown radius with an archived tape). *There is a deterministic algorithm that requires neither a distance bound nor the source slack D and has the following guarantee. Suppose a strongly r -satisfiable k -CNF on n variables has a strongly satisfying assignment at distance $h \geq 1$ from the initial assignment. Put $\ell_h = 1 + \lceil \log_2 h \rceil$ and $L = h + \log D + 1$. For every $D \geq 1$, $s \in [s_*, 1]$, and $\nu \in \mathcal{Q}_{s,D}$, the algorithm's total number σ of variable updates satisfies, for every integer $T \geq 0$,*

$$\mathbb{P}_\nu\{\sigma > T\} \leq \min \left\{ 1, CD\lambda^h \exp\left(-\frac{cT}{h^2\ell_h^2}\right) \right\},$$

and

$$\mathbb{E}_\nu\sigma \leq C \min \left\{ \Psi_s(L), h^2\ell_h^2 L \right\}.$$

Here $c, C > 0$ depend only on k, r . The update count includes every flip in every simulation and every coordinate restored during a reset. The algorithm keeps at most $2 + \lceil \log_2 n \rceil$ simulated assignments and at most σ archived source symbols. Each simulation may use any deterministic clause selector based only on its own local history. Work to find a violated clause is separate.

The algorithm. Stop immediately if the initial assignment x_0 satisfies the formula. Let $J = \lceil \log_2 n \rceil$. Keep one base simulation of the original walk and auxiliary simulations A_j , $0 \leq j \leq J$, of the Hamming-ball restart rule with radius parameter $H_j = 2^j$. All simulations start from x_0 and use the same source tape $\omega_1, \omega_2, \dots$; an auxiliary simulation can be initialized when first scheduled. Partition the nonnegative indices into groups

$$G_b = \{2^b - 1, \dots, 2^{b+1} - 2\}, \quad b \geq 0.$$

In round m , first select the base walk's clause using its past history; then read and archive ω_m and perform the indexed flip. Then choose the group G_b with $b = v_2(m)$, the largest integer with 2^b dividing m , and select its next index in cyclic order, beginning at its smallest index. If the selected index is at most J , advance that auxiliary simulation once, including its prescribed reset if needed. Otherwise leave the auxiliary slot unused. On its q th activation, an auxiliary simulation uses the archived symbol ω_q . Halt as soon as any simulation satisfies the formula.

Every auxiliary activation has $q \leq m$, so the required symbol is already available. Before its q th flip, its clause selector may use only that simulation's own states and $\omega_1, \dots, \omega_{q-1}$; it must not inspect ω_q or any later archived symbol. For example, every simulation can always choose the first violated clause in a fixed ordering. Full assignment arrays require $O(n \log(n+2))$ walk-state storage in addition to the archive and any selector workspace. Simulations may instead be initialized lazily relative to x_0 .

Proof of Theorem B.4. Define counterfactual complete simulations on the full tape, even if the combined algorithm halts earlier. Let τ_{base} be the base walk's local stopping time and let τ_j be that of A_j , counting its source-driven flips. Each simulation reads the original tape prefix in its original order. Its tape law is therefore exactly ν , so Theorems 4.1 and 4.4 apply to the respective simulations. Independence between them is not needed.

Let $N_j(M)$ be the number of activations assigned to A_j in the first M rounds. If $j \in G_b$, put $z = j - (2^b - 1)$ and $q_j = 2^{b+1}$. The activation rounds of this index are exactly

$$m \equiv 2^b(1 + 2z) \pmod{q_j}, \quad q_j \leq 2(j+1)^2.$$

Since the residue lies strictly between zero and q_j ,

$$N_j(M) \geq \lfloor M/q_j \rfloor \geq M/q_j - 1. \quad (31)$$

By round M the algorithm has made at most M base flips and M auxiliary source-driven flips. In each auxiliary simulation, the number of restored coordinates over all resets is at most the number of its source-driven flips: each restored coordinate was flipped during that attempt, and attempts are disjoint. The total number of variable updates by round M , or by earlier termination, is thus at most $3M$. In particular, $\sigma \leq 3\tau_{\text{base}}$ pathwise. Theorem 4.1 therefore gives $\mathbb{E}_\nu \sigma \leq C\Psi_s(h + \log D + 1)$.

Choose $j_* = \lceil \log_2 h \rceil$. Then $j_* \leq J$ and $h \leq H_{j_*} < 2h$. If $\sigma > T$, the combined algorithm has not terminated within the first $M = \lfloor T/3 \rfloor$ rounds, since termination in those rounds would use at most $3M \leq T$ updates. Consequently

$$\{\sigma > T\} \subseteq \{\tau_{j_*} > N_{j_*}(\lfloor T/3 \rfloor)\}.$$

Write c_0, C_0 for the constants in Theorem 4.4. From (31), $N_{j_*}(\lfloor T/3 \rfloor) \geq T/(3q_{j_*}) - 2$. Applying the known-radius bound to this one simulation gives

$$\begin{aligned} \mathbb{P}_\nu\{\sigma > T\} &\leq C_0 e^{2c_0} D \lambda^h \exp\left(-\frac{c_0 T}{3q_{j_*} H_{j_*}^2}\right) \\ &\leq C_0 e^{2c_0} D \lambda^h \exp\left(-\frac{c_0 T}{24h^2(j_* + 1)^2}\right). \end{aligned}$$

This proves the claimed tail after capping by one. Summing that capped exponential gives the second expectation bound. Taking the smaller of the two expectation estimates proves the result. Finally, each round reads one new symbol and performs at least one base update before any auxiliary work. Hence at termination the archive contains at most σ symbols. $\square$

C Proofs for restart policies

C.1 Necessity in Theorem 5.1

The proof in the main text establishes (i) $\Rightarrow$ (ii), and (ii) $\Rightarrow$ (iii) is immediate. Suppose now that $\alpha_s(x) = 1$. Exact prefix capacity gives a probability law $\mu \in \mathcal{Q}_{s,1}$ supported on $\mathcal{N}_x$. Fix an arbitrary deterministic restart policy. At each restart, begin an independent stream with law μ . This defines an adversarial source for the entire execution, not an assumption about the algorithm's access to independent randomness.

For any observed word w , the policy determines a unique segmentation into completed attempt words $b_1, \dots, b_j$ and an unfinished attempt word v . The resulting source therefore satisfies

$$\nu[w] = \mu[b_1] \cdots \mu[b_j] \mu[v] \leq P[b_1]^s \cdots P[b_j]^s P[v]^s = P[w]^s.$$

The formula is consistent on the prefix tree, including at restart boundaries. Every attempt follows an original nonterminating execution until the policy abandons it. Every prefix on which an attempt terminates has probability zero. The countable union of these prefixes still has probability zero, proving the converse and the stronger final assertion.

For completeness, the time bound in the sufficiency proof uses the horizon T , rather than the longest failed-attempt word: a successful path can avoid $\mathcal{B}$ and run longer than every word in that cover. If the longest cover word has length L , the slightly sharper bound $\Pr_\nu\{\tau_{\text{restart}} > T + kL\} \leq Dc^k$ also holds.

C.2 Proof of Theorem 5.2

The greatest feasible vector. The finite-horizon controlled recursion of Proposition A.2 is

$$V_0 = \mathbf{1}, \quad V_{t+1}(y) = \min \left\{ 1, \min_{f \in F_y} \sum_{z \in X} R_f(y, z) V_t(z) \right\}.$$

The sequence decreases, and its limit V is a fixed point because the state and action sets are finite. Every feasible vector v in (16) satisfies $v \leq V_t$ by induction, hence $v \leq V$. Conversely, the fixed point V is feasible. Thus V is the componentwise greatest feasible vector and the unique maximizer of the sum objective.

For every fixed history-dependent selector Λ , the finite-horizon optimality statement gives $\mathcal{C}_s(\{\tau_\Lambda > t\}) \geq V_t(x)$. Continuity in Theorem 2.1 therefore gives

$$\mathcal{C}_s(\{\tau_\Lambda = \infty\}) \geq V(x). \quad (32)$$

This argument fixes Λ before taking the decreasing-event limit; it does not interchange an infimum over selectors with that limit.

Selecting a stationary rule. Set $U = \{y : V(y) < 1\}$ and $S = X \setminus U$. For $y \in U$, let

$$I_y = \left\{ f \in F_y : \sum_{z \in X} R_f(y, z) V(z) = V(y) \right\}.$$

Each I_y is nonempty. On U , solve the second linear program

$$\max \left\{ \sum_{y \in U} h(y) : h \geq 0, \quad h(y) \leq 1 + \sum_{z \in U} R_f(y, z) h(z) \text{ for every } y \in U, f \in I_y \right\}. \quad (33)$$

If U is empty, this construction is vacuous and the selector may be chosen arbitrarily.

The second program is feasible and bounded. Indeed, a nonzero recession direction would be a vector $w \geq 0$ satisfying

$$w(y) \leq \sum_{z \in U} R_f(y, z) w(z) \quad (y \in U, f \in I_y).$$

Extend w by zero on S . For sufficiently small $\epsilon > 0$, $V + \epsilon w$ remains below $\mathbf{1}$ on U and preserves every constraint of (16): tight constraints on U use the displayed inequalities, nontight constraints have positive slack, and constraints on S can only be relaxed. This contradicts the componentwise maximality of V . Thus there is no nonzero recession direction, and the feasible polyhedron is bounded.

At an optimizer h , every state $y \in U$ has a binding action in (33). Otherwise increasing $h(y)$ slightly preserves its finitely many strict inequalities and relaxes all inequalities belonging to other states, contradicting optimality. Choose one binding action $\mu(y) \in I_y$, and choose μ arbitrarily on S . Define

$$A(y, z) = R_{\mu(y)}(y, z) \quad (y, z \in U), \quad b(y) = \sum_{z \in S} R_{\mu(y)}(y, z).$$

The binding constraints and membership in I_y give

$$h = \mathbf{1} + Ah, \quad V_U = AV_U + b. \quad (34)$$

In particular $h \geq 1$ and $Ah < h$, so $\rho(A) < 1$. Consequently,

$$h = (I - A)^{-1}\mathbf{1}, \quad V_U = (I - A)^{-1}b.$$

This second program is needed only to resolve ties properly. For example, at a state of value zero, both an update that always terminates and an update that always returns to the same state are tight at V ; the latter is not a valid choice for synthesis.

Exact optimality without restarts. Let W be the nontermination capacity vector of the fixed stationary selector μ . Equivalently, W is the greatest feasible vector for the fixed-matrix constraints $0 \leq w \leq \mathbf{1}$, $w \leq R_\mu w$, by the same decreasing-recursion argument. Since V is feasible for these constraints, $W \geq V$ and hence $W_S = \mathbf{1}$. On U ,

$$W_U \leq AW_U + b, \quad W_U \leq (I - A)^{-1}b = V_U.$$

Thus $W = V$. Combining this with (32) and the exact capacity identity proves that μ attains the stated minimum, simultaneously for every initial state and every $D \geq 1$.

Restart tail bound. Suppose $x_0 \in U$. Under μ and immediate restart on entry to S , the powered live matrix on U is

$$B = A + be_{x_0}^\top.$$

Set

$$g = h + \frac{h(x_0)}{1 - V(x_0)}V_U, \quad H = \max_{y \in U} g(y).$$

Equation (34) gives $Bg = g - \mathbf{1}$. Therefore $g \geq \mathbf{1}$, $\rho(B) < 1$, and $g = (I - B)^{-1}\mathbf{1}$. If $H > 1$, then $Bg \leq (1 - H^{-1})g$; if $H = 1$, then $B\mathbf{1} = 0$ and termination occurs after one source-consuming step.

Summing powered probabilities over complete surviving words gives

$$\mathbb{P}_\nu\{\tau_{\text{restart}} > t\} \leq \min\{1, De_{x_0}^\top B^t \mathbf{1}\}, \quad (35)$$

$$\mathbb{E}_\nu \tau_{\text{restart}} \leq Dg(x_0) = D \frac{h(x_0)}{1 - V(x_0)} = D \frac{e_{x_0}^\top (I - A)^{-1} \mathbf{1}}{1 - V(x_0)}. \quad (36)$$

For $H > 1$, the first bound is at most $Dg(x_0)(1 - H^{-1})^t$. The mean bound follows by summing the matrix bound over $t \geq 0$. These estimates invoke the source constraint only on whole execution prefixes; restarting does not restore a conditional source guarantee.

Impossibility for arbitrary controlled restart policies. Suppose $V(x_0) = 1$, and fix any deterministic nonanticipating policy that chooses updates and may restart to x_0 between successive steps. We construct one source in $\mathcal{Q}_{s,1}$ for this entire policy. Write $V = 0$ at terminated states. At a positive-probability history whose current live state is y and chosen update is f , set

$$L(y, f) = \sum_a \pi(a)^s V(\Phi_f(y, a)) \geq V(y) > 0, \quad q(a \mid \text{history}) = \frac{\pi(a)^s V(\Phi_f(y, a))}{L(y, f)}.$$

These probabilities sum to one. They assign zero probability to termination and to states of value zero. Within any attempt, their product along a word b ending at z is at most $P[b]^s V(z)/V(x_0) = P[b]^s V(z)$, by telescoping the successive values and using $L(y, f) \geq V(y)$.

For any whole execution prefix w , the policy determines its unique segmentation into completed attempts and a current partial attempt. Multiplying the preceding bounds over these segments

gives $\nu[w] \leq P[w]^s$, since each endpoint value is at most one. Thus the constructed law lies in $\mathcal{Q}_{s,1}$ and has nontermination probability one. Its definition may depend on the fixed policy, as permitted by the minimax quantifiers. This proves necessity even for policies that retain their entire histories across restarts.

Exact computation. If the powered coefficients $R_f(y, z)$ are rational, the first linear program, its unique optimizer, the tight action sets, and the second linear program all have polynomial-size rational descriptions. Exact linear programming and rational linear algebra therefore compute μ , S , and g in polynomial bit complexity in the explicit state-action input. The resulting execution-time constants need not be polynomially bounded. Approximate representations of arbitrary real values $\pi(a)^s$ do not by themselves provide the exact rational input required by this assertion.

C.3 Proof of Proposition 5.3

Use source alphabet $\{0, 1, 2, 3\}$, initial state x , and the transition table below, where $\dagger$ denotes termination.

	0	1	2	3
x	$\dagger$	y	z	z
y	$\dagger$	y	y	y
z	z	$\dagger$	$\dagger$	$\dagger$

Under the uniform product law,

$$\mathbb{E}_P \tau = 1 + \frac{1}{4} \cdot 4 + \frac{1}{2} \cdot \frac{4}{3} = \frac{8}{3}.$$

At exponent $1/2$, each symbol has weight $1/2$. The original powered live matrix, in state order (x, y, z) , is

$$R = \begin{pmatrix} 0 & 1/2 & 1 \\ 0 & 3/2 & 0 \\ 0 & 0 & 1/2 \end{pmatrix}.$$

For every $T \geq 1$, the failed length- T words F_T have total powered mass

$$Z_T^{\text{live}} := \sum_{w \in F_T} P[w]^{1/2} = \frac{1}{2}(3/2)^{T-1} + (1/2)^{T-1} \geq 1.$$

For $T = 1, 2$, the values are $3/2, 5/4$; for $T \geq 3$, the first term alone exceeds one. The original nontermination capacities are $(1/2, 1, 0)$.

For a finite cutoff T , put the probability law $q_T(w) = P[w]^{1/2}/Z_T^{\text{live}}$ on F_T . At a nonempty partial word u of length $j \leq T$, the ratio $q_T[u]/P[u]^{1/2}$ is zero or one of

$$\frac{(3/2)^{T-j}}{Z_T^{\text{live}}} \leq 2 \quad \text{on the } y \text{ branch}, \quad \frac{(1/2)^{T-j}}{Z_T^{\text{live}}} \leq 1 \quad \text{on the } z \text{ branch}.$$

At a complete block, this ratio is $1/Z_T^{\text{live}} \leq 1$. For any predetermined finite-cutoff sequence, concatenate independent blocks with these laws. Each complete block contributes a ratio at most one, and the final partial block contributes at most two. The resulting law $\mu \in \mathcal{Q}_{1/2,2}$ is supported on nontermination of the scheduled procedure. If the schedule has a final infinite cutoff, choose its first symbol to be 1, then choose independent uniform symbols from $\{1, 2, 3\}$ forever. The partial-prefix ratios in this final attempt are also at most two. Hence $\mu/2$ is a subprobability measure satisfying the unit prefix caps. Prefix-flow completion extends it to a probability law in

$\mathcal{Q}_{1/2,1}$ while retaining nontermination mass at least $1/2$. The adversarial source may depend on the predetermined schedule.

The adaptive rule restarts immediately on entry to y . Its powered live matrix on (x, z) is $B = \begin{pmatrix} 1/2 & 1 \\ 0 & 1/2 \end{pmatrix}$. The exact capped recursion gives $v_T(z) = 2^{-T}$ and $v_0(x) = v_1(x) = v_2(x) = 1$. For $T \geq 3$,

$$v_T(x) = \frac{1}{2}v_{T-1}(x) + 2^{-(T-1)} = T2^{1-T}.$$

The formula also holds at $T = 2$. Exact prefix capacity now gives $\max_{\nu \in \mathcal{Q}_{1/2,D}} \Pr_{\nu}\{\tau_{\text{adaptive}} > T\} = \min\{1, DT2^{1-T}\}$.

D Proofs for scheduling

D.1 Optimal priority for heterogeneous retry tasks

Proof of Theorem 3.1. For a set S of unfinished tasks, the controlled recursion gives

$$V_{t+1}(S) = \min \left\{ 1, \min_{i \in S} ((W - w_i)V_t(S) + w_i V_t(S \setminus \{i\})) \right\}, \quad (37)$$

with $V_t(\emptyset) = 0$ and $V_0(S) = 1$ for nonempty S . Symbols with the same success status lead to the same state, so their powered masses add even when a selector remembers the individual labels. Proposition A.2 gives the exact optimum $\min\{1, DV_T(S)\}$.

We first record three monotonicities. Values decrease with t , because the recursion operator is monotone and $V_1 \leq V_0$. They increase when tasks are added. Indeed, suppose $R \subseteq S$ and argue by induction on t . An action $i \in R$ has at least its corresponding cost on R . An action $i \in S \setminus R$ has cost at least $WV_t(R) \geq V_t(R) \geq V_{t+1}(R)$, since $W \geq 1$. Taking the minimum and truncating at one preserve the comparison.

For fixed W , increasing a task's weight cannot increase the value. To see this, compare recursions in which its weight is a and $b \geq a$. By induction, the present-state values satisfy $X_a \geq X_b$, and the common value after completing that task is $Y \leq X_b$. The corresponding action costs differ by

$$(W - a)X_a + aY - ((W - b)X_b + bY) = (W - a)(X_a - X_b) + (b - a)(X_b - Y) \geq 0.$$

Other action costs are ordered by induction. These arguments apply to the abstract recursion for all $0 < w_i \leq W$ and $W \geq 1$, without requiring a modification of the reference source.

Subtracting the costs of two possible first tasks cancels the common W term. What remains is a comparison of the values obtained by removing either task. This motivates the following stronger invariant, which we prove by induction on t : for distinct $i, j \in S$ with $a = w_i < b = w_j$,

$$(b - a)V_t(S) \leq bV_t(S \setminus \{j\}) - aV_t(S \setminus \{i\}). \quad (38)$$

It holds at $t = 0$. The inequality says exactly that the action for i has no larger cost than the action for j in (37). Equal-weight tasks are interchangeable. Thus, if the invariant holds at time t , a minimum-weight task is an optimal first action for horizon $t + 1$.

We shall use the following elementary truncation fact. If $0 \leq z \leq y \leq x$ and $(b - a)x \leq by - az$, then

$$(b - a) \min\{1, x\} \leq b \min\{1, y\} - a \min\{1, z\}. \quad (39)$$

If $x \leq 1$, nothing changes. If $y \geq 1$, the right side is at least $b - a$. Otherwise $x > 1 > y$, and the unchanged right side is at least $(b - a)x \geq b - a$.

Fix i, j as above, and choose a minimum-weight task $k \in S$. First suppose $k \notin \{i, j\}$ and put $c = w_k$. The optimal first action in S , $S \setminus \{i\}$, and $S \setminus \{j\}$ can be k . Before truncation, each value is the same nonnegative linear combination, with coefficients $W - c, c$, of the corresponding values with and without k at time t . Combine (38) for S and $S \setminus \{k\}$ to obtain the desired inequality before truncation. The three quantities have order $x \geq y \geq z$: adding a task increases the value, and replacing a weight a by b decreases it. Equation (39) proves the invariant after truncation.

The remaining case is $k = i$, since $a < b$ excludes $k = j$. Write

$$X = V_t(S), \quad Y = V_t(S \setminus \{j\}), \quad Z = V_t(S \setminus \{i\}), \quad H = V_t(S \setminus \{i, j\}).$$

The monotonicities give $X \geq Y \geq Z \geq H$. Set

$$x = (W - a)X + aZ, \quad y = (W - a)Y + aH, \quad z = (W - b)Z + bH.$$

The first two quantities are the optimal costs before truncation for S and $S \setminus \{j\}$. The third is a permissible cost for $S \setminus \{i\}$, obtained by choosing j . Hence its truncated value is an upper bound on $V_{t+1}(S \setminus \{i\})$. We have $x \geq y$ and

$$y - z = (W - a)(Y - Z) + (b - a)(Z - H) \geq 0.$$

Furthermore,

$$by - az - (b - a)x = (W - a)(bY - aZ - (b - a)X) \geq 0.$$

Apply (39), then replace its last truncated quantity by the no-larger actual value. This proves (38) at $t + 1$ and completes the induction. Thus increasing-weight priority is optimal at every horizon. Following that order leaves only suffixes of the sorted list and gives the stated $O(nT)$ recursion. Monotonicity of $x \mapsto \min\{1, Dx\}$ proves simultaneous optimality for every $D \geq 1$.

Finally, fix $q > 0$, let $W_q = \sum_a \pi(a)^q$, and consider the tilted law $\pi_q(a) = \pi(a)^q / W_q$. Under its independent stream, preassign independent streams to the tasks. The number of attempts for task i is geometric with success probability $\sum_{a \in B_i} \pi_q(a)$. The workloads are independent and finite almost surely. Every permitted selector uses exactly their sum: before all tasks complete it must attempt some unfinished task, and none can receive more than its preassigned workload. The next unrevealed symbol of any selected stream has law π_q , even when the selection uses all previous observations. The stopping-time distribution is therefore selector independent. Multiplication by W_q^T proves equality of surviving-prefix power sums. The identity $Z_T(q) = W_q \tilde{Z}_{T-1}(q)$ proves the same claim for the completed-step convention. $\square$

D.2 Equal stopping laws and different robustness

Proof of Theorem 3.2. Put

$$u = \sqrt{\pi(0)} = \frac{\sqrt{15}}{4}, \quad v = \sqrt{\pi(1)} = \frac{1}{4}.$$

We first construct ν_n . Through time $8n$, generate actions with probabilities $3/4$ for 0 and $1/4$ for 1 until the n th zero appears; after that zero, output only zeros through time $8n$. After time $8n$, generate independent actions according to P . This specifies one law on tapes, without reference to the selector.

A prefix of length at most $8n$ preceding the n th zero, with k zeros, satisfies

$$\frac{\nu_n[w]}{P[w]^{1/2}} = \left(\frac{3}{\sqrt{15}} \right)^k \leq 1.$$

Suppose the n th zero appears at time $L \leq 8n$. For a positive-mass prefix of length $t \in [L, 8n]$, the corresponding ratio is

$$\left(\frac{3}{\sqrt{15}}\right)^n u^{-(t-L)} \leq \left(\frac{3/4}{u^8}\right)^n < 1,$$

since $L \geq n$ and $u^8 = (15/16)^4 > 3/4$. Zero-mass prefixes satisfy the required bound automatically. After time $8n$, each new reference action multiplies the ratio by $\pi(a)^{1/2} \leq 1$. Thus every finite prefix satisfies the defining inequality for $\mathcal{Q}$. Under Λ_A , either fewer than n zeros have appeared by time $8n$, so an A task remains unfinished, or the B task is attempted only after the n th zero and subsequently sees zeros alone. In both cases $\tau > 8n$, proving (8).

For (9), consider an arbitrary $\nu \in \mathcal{Q}$ and use Λ_B . If no 1 appears during the first $4n$ actions, the tape begins with 0^{4n} , an event of ν -probability at most

$$u^{4n} = \left(\frac{225}{256}\right)^n.$$

Otherwise let $j \leq 4n$ be the first position containing 1. This completes B , and survival after $8n$ repairs requires fewer than n zeros among the remaining $\ell = 8n - j \geq 4n$ actions. Set $r = v/(3u) = 1/(3\sqrt{15})$. Their total power weight is at most

$$\begin{aligned} \sum_{k=0}^{n-1} \binom{\ell}{k} u^k v^{\ell-k} &\leq r^{-(n-1)} (v + ur)^\ell \\ &\leq r^{-(n-1)} (1/3)^{4n} = r \left(\frac{\sqrt{15}}{27}\right)^n. \end{aligned}$$

The first-1 prefix $0^{j-1}1$ has power weight $u^{j-1}v$. Applying the cylinder bound and summing over $j \leq 4n$ gives the claimed second term because

$$\sum_{j=1}^{4n} u^{j-1}v \leq \frac{v}{1-u} = 4 + \sqrt{15}.$$

This argument holds for every source in $\mathcal{Q}$, including the source ν_n used above. To obtain the simpler constant in (9), put $\alpha = \sqrt{15}/27$, $\beta = 225/256$, and $c = (4 + \sqrt{15})/(3\sqrt{15})$. The bound is $\beta^n + c\alpha^n$. Since $\alpha < \beta$ and $c\alpha/\beta = 256(4 + \sqrt{15})/18225 < 1/8$, it is at most $(9/8)\beta^n < 1$ for every $n \geq 1$.

For the lower bound in (9), take an i.i.d. source with probabilities u and $1 - u$. Both are at most the corresponding square-root reference probabilities, since $1 - u \leq v$. It belongs to $\mathcal{Q}$, and its prefix 0^{8n} leaves B unfinished under every selector. Its probability $u^{8n} = (15/16)^{4n}$ is therefore a lower bound for every selector, including the minimax optimum.

The selector invariance of the stopping law and all prefix power sums follows from Theorem 3.1, applied to these success sets. $\square$

E Proof for noisy predecessor search

Proof of Theorem 6.1. Set $x_0 = -\infty$ and $x_{n+1} = +\infty$. Leaf i represents $[x_i, x_{i+1})$ and returns index i , with zero meaning no predecessor. Internal nodes represent intervals $[x_i, x_j)$ and split at x_k , $i < k < j$. Chain nodes inherit their leaf's interval and label. These half-open intervals handle equality using strict queries.

At a nonterminal state with interval $[x_i, x_j)$, use the lower and upper comparisons $x < x_i$ and $x < x_j$ when their boundaries are finite. Verification requires the lower answer to be false and the upper answer to be true. An infinite boundary is known exactly; replace its comparison by a dummy comparison $x < x_1$ and ignore the answer. At an original internal node the third comparison is its split comparison $x < x_k$; at an original leaf or chain node it is another dummy $x < x_1$. Each of these three logical comparisons is answered by the majority of exactly ℓ repetitions of the same primitive query. All 3ℓ queries are made before the move, including dummy queries and queries after an earlier majority already makes verification fail. On failure move to the parent, with a requested upward move at the root replaced by staying there. On success take the split-selected child at an internal node and the unique downward chain edge otherwise. At a chain endpoint return immediately, without further queries.

Let z be the correct original leaf and z^* its chain endpoint. If a node's interval excludes x , three correct majorities make verification fail, and its parent is the neighbor toward z^* . If its interval contains x , correct answers select the child toward z^* , or advance on z 's chain. Thus every raw block in G moves one edge toward z^* . Every other block moves at most one edge. For live states use the positive integer potential $d(v) = \text{dist}(v, z^*)$, initially $d_0 = 2h + L$; assign potential zero on termination.

For odd ℓ , define the powered mass of a correct majority by

$$M_s = \sum_{j > \ell/2} \binom{\ell}{j} p^{sj} (1-p)^{s(\ell-j)}.$$

Since $p > 1/2$, the law of large numbers ensures an odd ℓ with $M_1 > 2^{-1/3}$. Choose the least such ℓ , so it depends only on p . Group raw errors into consecutive blocks of length 3ℓ , with reference mass $\pi(e) = p^{3\ell - |e|_1} (1-p)^{|e|_1}$. Equation (17) directly bounds every length- T block prefix by D times its reference mass to power s ; no change in the primitive source assumption is made. Summing over raw patterns gives

$$a_s = M_s^3, \quad b_s = (p^s + (1-p)^s)^{3\ell} - M_s^3.$$

Both are sums of $\pi(e)^s$ over fixed nonempty sets, so both strictly decrease in s . At $s = 0$, their product is $7 \cdot 2^{6\ell-6} > 1/4$; at $s = 1$, it is $M_1^3(1 - M_1^3) < 1/4$. This proves existence and uniqueness of s_c . Also $a_{s_c} \geq M_1^3 > 1/2$, so $a_{s_c} = \lambda/2$, $b_{s_c} = 1/(2\lambda)$, and $\lambda > 1$. This standard amplification is analyzed on raw patterns: the correct powered mass is M_s^3 , not M_1^{3s} . No source condition on the majority answers is assumed.

For every nonnegative increasing f with $f(0) = 0$, the live children of a history of potential d satisfy

$$\sum_{e: \text{live}} \pi(e)^{s_c} f(d_e) \leq a_{s_c} f(d-1) + b_{s_c} f(d+1). \quad (40)$$

This is the complete one-step input to the proof of Lemma B.1: its tree induction uses powered edge masses, so the same proof applies to these nonuniform edges. In particular, for any integer $m > d_0$, the powered mass of prefixes first reaching m before termination is at most

$$H_m(d_0) = \frac{d_0}{m} \lambda^{d_0-m}, \quad H_m(d) = \frac{d}{m} \lambda^{d-m}.$$

Indeed, $H_m(0) = 0$, $H_m(m) = 1$, and $a_{s_c} H_m(d-1) + b_{s_c} H_m(d+1) = H_m(d)$; expanding the finite tree to any depth bounds its first-hit prefixes by $H_m(d_0)$, and increasing the depth gives the infinite-horizon bound. The remaining live prefixes confined below m have powered mass at most

$$\frac{\lambda^{d_0}}{\sqrt{\lambda^2 - 1}} \exp\left(-\frac{\pi^2 T}{2(2m-1)^2}\right),$$

by the same reflected-strip comparison and diagonal conjugation in that lemma. Multiplying the two cover sums by D and optimizing m therefore gives

$$\Pr_{\nu}\{Q > 3\ell T\} \leq \min\{1, C_p D \lambda^{d_0} e^{-c_p T^{1/3}}\}, \quad \mathbb{E}_{\nu} Q \leq C_p (d_0 + \log D + 1)^3.$$

The inclusion of source classes gives this bound for all $s \geq s_c$. For $s > s_c$, since $\max_e \pi(e) = p^{3\ell}$,

$$\frac{a_s}{\lambda} + b_s \lambda \leq p^{3\ell(s-s_c)} \left(\frac{a_{s_c}}{\lambda} + b_{s_c} \lambda \right) = p^{3\ell(s-s_c)}.$$

Finite-tree induction on the powered sum of λ^d over live depth- T prefixes yields $\Pr_{\nu}\{Q > 3\ell T\} \leq \min\{1, D \lambda^{d_0} p^{3\ell(s-s_c)T}\}$. Summing and counting 3ℓ queries per move gives $C_p(d_0 + \log D + 1)/(s - s_c)$. Since $d_0 = 2h + L = O_p(K)$, these bounds prove the query claim and almost-sure termination.

Finally consider an incorrect leaf u of depth d_u , and let c be the depth of its lowest common ancestor with z . Its chain endpoint u^* satisfies

$$\text{dist}(u^*, z^*) = 2h + 2d_u - 2c + 2L \geq d_0 + L + 2.$$

Set $m = d_0 + L + 1$. Any run terminating at u^* must first reach distance m at a live state: its final endpoint has distance at least $m + 1$, and each move changes distance by at most one. Consequently the first-hit bound, applied before resetting the terminal potential to zero, gives

$$\Pr_{\nu}\{\text{incorrect predecessor}\} \leq D H_m(d_0) \leq D \lambda^{-L-1} \leq \eta.$$

Only the algorithm's visited leaf label is returned; the unknown correct endpoint is used solely in this analysis. $\square$

F Composition under a shared source

Sharing a random stream can alter robustness even when a selector may interleave components arbitrarily. The following process gives an exact comparison between a global prefix guarantee and separate component guarantees.

Each component is a connected N -vertex graph with binary vertex values. Select an edge whose endpoints disagree. Four source labels replace its endpoint values by 00, leave them unchanged, reverse them, or replace them by 11. The reference probabilities are $(49/100, 1/4, 1/4, 1/100)$, respectively. Constant components are finished and never selected again. This extends the discordant voting model of Cooper, Dyer, Frieze, and Rivera [CDFR18].

Let $\Delta_N \in (1/2, 1)$ be the unique solution of

$$2 \cdot 4^{-\Delta_N} + 2(7/100)^{\Delta_N} \cos(\pi/N) = 1.$$

Corollary F.1 (Small individual failure, certain joint failure). *For every $N \geq 3$, start each of $N - 1$ components with exactly one vertex equal to one and all others zero. In isolation, each has worst-case nontermination probability*

$$\max_{\nu \in \mathcal{Q}_{\Delta_N, 1}} \nu\{\tau_i = \infty\} = 7^{-\Delta_N(N-2)} = \exp[-\Theta(N)].$$

There is one source in $\mathcal{Q}_{\Delta_N, 1}$ under which their combined execution never terminates, simultaneously for every deterministic nonanticipating rule that chooses a disagreeing edge. The rule may interleave components arbitrarily and remember all past labels. Under the independent reference source, every such rule has expected total running time at most $25(N - 1)/12$.

The following formula proves the corollary and also allows components of different sizes and arbitrary initial assignments.

Theorem F.2 (Exact robustness under arbitrary interleaving). *Fix $N \geq 3$, put $s = \Delta_N$, $\theta = \pi/N$, and $r = 7^s$, and define*

$$V_N(k) = \begin{cases} 0, & k = 0, \\ r^{k-N+1} \frac{\sin(k\theta)}{\sin\theta}, & 1 \leq k < N-1, \\ 1, & k \geq N-1. \end{cases}$$

Let the graph be a disjoint union of finitely many connected components, each with at least N vertices. Let K_0 count the ones in the initially nonconstant components. For every deterministic nonanticipating edge selector and every $D \geq 1$,

$$\max_{\nu \in \mathcal{Q}_{s,D}} \nu\{\tau = \infty\} = \min\{1, DV_N(K_0)\}. \quad (41)$$

One source depending only on N, K_0, D attains this value simultaneously for every allowed graph, initial assignment with count K_0 , and selector. Under the independent reference source,

$$\mathbb{E}_P \tau \leq \frac{25}{12} K_0. \quad (42)$$

Proof. Write $V = V_N$ and let

$$d = (49/100)^s, \quad b = 2(1/4)^s, \quad u = (1/100)^s, \quad a = \sqrt{du} = (7/100)^s.$$

The equation defining s is $b + 2a \cos \theta = 1$. Its left side is strictly decreasing in s , exceeds one at $s = 1/2$, and is below one at $s = 1$. Thus $1/2 < s < 1$. In particular, $b < 1$, $d = ar$, $u = a/r$, and $r > \sqrt{7} > 2 \cos \theta$.

Set $h(k) = r^k \sin(k\theta)$ for $0 \leq k \leq N$. The sine addition formula gives

$$dh(k-1) + bh(k) + uh(k+1) = h(k), \quad 1 \leq k < N.$$

Moreover, h is strictly increasing on $1, \dots, N-1$: for $1 \leq k \leq N-2$, $\sin(k\theta) \leq 2 \cos \theta \sin((k+1)\theta)$ and $r > 2 \cos \theta$. Hence $0 < V(k) < 1$ for $1 \leq k < N-1$, and the same harmonic identity holds for V there. At the upper boundary it gives

$$dV(N-2) + b = 1. \quad (43)$$

At any history, let K count the ones in components that are still nonconstant. A downward action sends K to $K-1$; either neutral action leaves it unchanged. An upward action sends K to $K+1$, unless it makes an L -vertex component all ones, in which case K becomes $K+1-L$. Such a completion requires $K \geq L-1 \geq N-1$. Thus below $N-1$ the count has the same increments for every selector, even though this need not hold above that level.

For the upper bound, only $1 \leq K_0 < N-1$ requires proof. Cover each nonterminating tape at its first visit to $K = N-1$, or at depth T if it has not yet visited that level. Before the first visit, the powered count matrix on $1, \dots, N-2$ is tridiagonal with lower diagonal d , diagonal b , and upper diagonal u . Call it A . Diagonal similarity to a symmetric tridiagonal matrix gives

$$\rho(A) = b + 2a \cos \frac{\pi}{N-1} < 1.$$

The harmonic identity with boundary values $V(0) = 0$ and $V(N-1) = 1$ therefore makes the total powered mass of first visits to $N-1$ equal to $V(K_0)$. The remaining live prefixes at depth T have

powered mass $e_{K_0}^\top A^T \mathbf{1}$, which tends to zero. Applying the prefix bound to this finite cover and letting $T \rightarrow \infty$ gives $\nu\{\tau = \infty\} \leq \min\{1, DV(K_0)\}$.

For the matching source, suppose $K_0 > 0$. An auxiliary law q maintains a counter $Y_0 = K_0$. Let $\delta(z) \in \{-1, 0, 1\}$ be the increment assigned to action label z . When $1 \leq Y_t < N - 1$, use

$$q(z \mid Y_t) = \pi(z)^s \frac{V(Y_t + \delta(z))}{V(Y_t)}.$$

When $Y_t \geq N - 1$, give each neutral label probability $(1/4)^s$, the downward label probability $1 - b$, and the upward label probability zero. Update $Y_{t+1} = Y_t + \delta(z)$. The harmonic identity normalizes these probabilities below $N - 1$, and normalization above that level is immediate. The downward probability at one is zero, so Y_t stays positive.

Every upward move starts below $N - 1$, so it cannot finish a component. A downward completion produces an all-zero component and has no additional effect on the count. Consequently $K_t = Y_t$ for every graph, initial assignment, and selector under consideration. Every tape in the support of q is nonterminating for all these processes.

Equation (43), together with $V(N - 2) < 1$, gives $1 - b = dV(N - 2) < d$. It follows in the upper range as well as the lower range that

$$q(z \mid Y_t) \leq \pi(z)^s \frac{V(Y_{t+1})}{V(Y_t)}.$$

Multiplying along a prefix yields

$$q[w] \leq P[w]^s \frac{V(Y_{|w|})}{V(K_0)} \leq \frac{P[w]^s}{V(K_0)}. \quad (44)$$

Put $p = \min\{1, DV(K_0)\}$. The subprobability law $\eta = pq$ has mass p and obeys $\eta[w] \leq DP[w]^s$. It extends to a probability law ν obeying the same bound: at each node, distribute its assigned mass among the children, preserving their η masses and respecting their capacities $DP[wz]^s$. This is possible because the lower bounds sum to $\eta[w]$, whereas the capacities sum to $DP[w]^s \sum_z \pi(z)^s \geq DP[w]^s$. Starting with root mass one and using a fixed label order gives an extension depending only on N, K_0, D . Its common nontermination mass is at least p , proving equality in (41). If $K_0 = 0$, all components are already constant and the assertion is immediate. When $K_0 \geq N - 1$ and $D = 1$, $p = 1$ and q itself is the required source.

Finally, under P , a live update has downward drift at least $49/100 - 1/100 = 12/25$ in K ; finishing an all-one component only increases that downward drift. Summing the conditional drift bound up to $\tau \wedge t$ gives $(12/25)\mathbb{E}_P(\tau \wedge t) \leq K_0$. Letting $t \rightarrow \infty$ proves (42). $\square$

For one component starting with one vertex equal to one, $V_N(1) = 7^{-\Delta_N(N-2)}$. For $N - 1$ such components, $V_N(N - 1) = 1$ and the same source forces nontermination under every interleaving. Equation (42) gives the reference mean bound. This proves Corollary F.1.

AI methodologies

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